

CLINOMA PLATFORM

PEDIATRICS

FLASHSPACE EDITION

Interactive High-Yield Visual Highlights & Explanations

EXCLUSIVE DIGITAL STUDY GUIDE

This document is digitally watermarked and registered for personal academic use only.

TABLE OF CONTENTS

CHAPTER / CLINICAL SLIDE	PAGE
I. GROWTH AND DEVELOPMENT	P. 3
Growth charts	p. 4
Development milestones during 1st 4 years	p. 5
Key warning signs & delayed milestones	p. 6
II. NUTRITION	P. 7
Advantages of breast feeding & contraindication	p. 8
Nutritional disorders: Kwashiorkor & Marasmus	p. 9
Nutritional disorders: Rickets	p. 10
III. GIT	P. 11
Cow's milk allergy in Pediatrics	p. 12
GERD & Hypertrophic Pyloric Stenosis	p. 13
Vomiting in pediatrics	p. 14
Abdominal pain in pediatrics	p. 15
IV. GENETIC DISEASES	P. 16
Chromosomal disorders: Down Syndrome	p. 17
Chromosomal disorders: Turner syndrome	p. 18
Prenatal diagnosis (importance, indications & types)	p. 19

CHAPTER / CLINICAL SLIDE	PAGE
V. ENDOCRINOLOGY	P. 20
Short stature	p. 21
Hypothyroidism	p. 22
Diabetes Mellitus Type 1 & DKA	p. 23
Childhood obesity	p. 24
VI. HEMATOLOGY AND ONCOLOGY	P. 25
Iron deficiency anaemia	p. 26
Aplastic anaemia	p. 27
Chronic hemolytic anaemia	p. 28
RBC abnormalities: Spherocytosis	p. 29
RBC abnormalities: Thalassemias	p. 30
RBC abnormalities: G6PD deficiency	p. 31
Hemophilia	p. 32
ITP (Immune Thrombocytopenic Purpura)	p. 33
ALL & Prognostic factors	p. 34
Hodgkin lymphoma	p. 35

CHAPTER / CLINICAL SLIDE	PAGE
VII. CVS	P. 36
Acyanotic Heart: VSD (Part 1)	p. 37
Acyanotic Heart: VSD (Part 2)	p. 38
Acyanotic Heart: ASD	p. 39
Acyanotic Heart: PDA	p. 40
Cyanotic Heart: TGA	p. 41
Cyanotic Heart: Fallot / TOF	p. 42
Heart Failure	p. 43
VIII. CNS	P. 44
Cerebral Palsy	p. 45
Floppy infant	p. 46
Duchenne muscle dystrophy	p. 47
Infection	p. 48

Tip: Use the interactive landscape view on your tablet or laptop for optimal clinical visualization.

CHAPTER 01

I. Growth and development



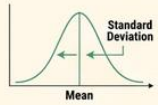
GROWTH CHARTS • DEVELOPMENT MILESTONES DURING 1ST 4 YEARS • KEY WARNING SIGNS & DELAYED MILESTONES

GROWTH CHARTS: DEFINITION, CONSTRUCTION & CLINICAL APPLICATION

DEFINITION & CONSTRUCTION



- Growth charts** consist of a **series of curves** illustrating the **distribution** of selected body measurements in **children**.



- Expressed in **percentiles** or as **standard deviation** from the mean for on average **average height or weight** for age.

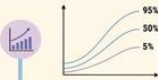


- Constructed by **observing the growth** of a **large number of normal children** over time.

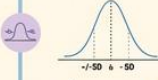


- Height, weight, and head circumference** of a child can be compared to **expected parameters** of same age & sex to determine if the child is **growing appropriately**.

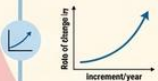
TYPES OF GROWTH CHARTS



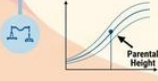
1. Percentile curves.



2. Standard deviation curves.



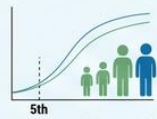
3. Velocity curves.



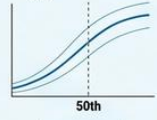
4. Conditional centiles.

IMPORTANCE AND INTERPRETATION / CLINICAL USES

1. Percentile Curves:



If height is on the **5th centile**, for every 100 children, **5%** would be shorter and **95%** taller (percentage chance of child value relative to a given centile).



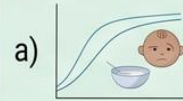
The **standard or reference curve** is attributed to the **50th percentile**.



Consists of: "**Length-for-age**," "**Weight-for-age**," "**Stature-for-age**," "**BMI-for-age**," "**Head circumference-for-age**," and "**Weight-for-length**."



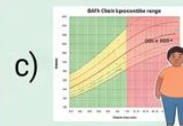
Clinical Uses:



- a) Early detection of **protein energy malnutrition** by following progression; **flattening** of the curve is an early sign of nutritional deficiencies.



- b) Early detection of **hydrocephalus** by following progression; **jumping** to above curves is an early sign of an enlarging head.

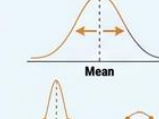


- c) Determination of **risk for obesity** if BMI lies between **85th - 95th percentiles**; if above **95th**, the infant is diagnosed as **overweight**.

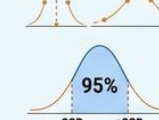


- d) Diagnosis of **nutritional diseases** by using **Wellcome** and **Waterlow** classifications.

2. Standard Deviation Curves:



Shows how much **variation** or '**dispersion**' there is from the **average (mean)**.



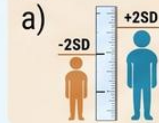
A **low standard deviation** indicates points **close** to the mean, while a **high standard deviation** indicates data **spread out** over a large range.



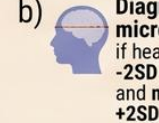
An SDS within **two standard deviations** encompasses about **95%** of healthy children.



Clinical Uses:



- a) **Diagnosis of short stature** if height is **-2SD** below the mean, and **tall stature** if height is **+2SD** above the mean.



- b) **Diagnosis of microcephaly** if head circumference is **-2SD** below the mean, and **macrocephaly** if **+2SD** above the mean.

3. Velocity Curves:



Considers the **change (increment)** in growth **over time** from year to year.



It can determine the **response to treatment** of **pathologic short stature**, such as **growth hormone** and **thyroxine**.

4. Conditional Centiles:



Centiles in which reference data are **conditional on** or **adjusted for some specific factor** over or above age and sex.



The child height percentile **conditional on midparent height** is important for the diagnosis of **familial short stature**.

DEVELOPMENTAL MILESTONES CHECKLIST: A VISUAL GUIDE (BY AGE)

BIRTH / NEWBORN

Gross motor: Flexion attitude when prone. Limbs flexed, symmetrical posture. **Marked head lag on pulling up.**



Fine motor: Blinks to flashing light.



1 MONTH

Gross motor: Raise head slightly when prone.



Fine motor: Respond to sound by blinking or startle.



Social: Prefers looking to face (regards face).

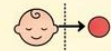


2 MONTHS / 6-8 WEEKS

Gross motor: Raises head more when prone briefly. Lifts head to 45 degrees in prone.



Fine motor: Follow moving objects passing midline.



Social: Smile socially.



3 MONTHS

Gross motor: Head support. Lifts head 45 degrees when prone for a sustained period.



Fine motor: Hand regards (brings hands together).



Language: Coos.



Social: Anticipate feeding.



4 MONTHS

Gross motor: Moro reflexes disappear. Sits with support. Rolls from back to front.



Fine motor: Turn head to sounds. Bring objects to mouth.



Language: Laughs loudly.



Social: Enjoys looking around. Recognize mother. Aware of strange faces.



6 MONTHS / 6-8 MONTHS

Gross motor: Sits without support at 6 months with a round back.



Fine motor: Transfers objects from one hand to another.



7 MONTHS / 8 MONTHS

Gross motor: Sit without support at 8 months with a straight back.



Fine motor: Grasp by palm.



9 MONTHS / 8-9 MONTHS / 10 MONTHS

Gross motor: Creeps. Crawling. Pulls to standing position. Cruises around furniture at 10 months.



Fine motor: Grasp by thumb and finger (pincer grip).



Language: Say dada & mama nonspecific.



Social: Explores.



12 MONTHS

Gross motor: Walks alone or with one hand. Walks unsteadily, broad gait, hands apart.



Fine motor: Stops mouthing of objects. Release objects on request.



Language: Say 2-3 words and understands several words.



Social: Imitates. Cooperates in dressing.



15 MONTHS

Gross motor: Walks well. Walks steadily.



Fine motor: Drinks from cup. Feeds self with spoon.



Language: Jargon (baby's own language).



Social: Sphincter control by daytime.



18 MONTHS

Gross motor: Runs.



Fine motor: Build a tower of 3 cups.



Language: Average 10 words.



Social: List-o-yaunme 10 words.



18 MONTHS

Gross motor: Runs.



Fine motor: Build a tower of 3 cups.



Language: Average 10 words.



2 YEARS

Gross motor: Upstairs with one hand. Downstairs step by step.



Fine motor: Draw a horizontal stroke on imitation.



Language: 2-3 words sentence.



Social: Group playing. Sphincter control day and night.



3 YEARS

Gross motor: Jumps.



Fine motor: Copies a circle, cross.



Language: Counts 3 objects.



Social: Dress himself with supervision.



4 YEARS

Gross motor: Hops on one foot.



Fine motor: Buttons & unbuttons. Draw a man.



Language: Say simple songs. Counts to 4.



Social: Goes to toilet. Names colors.



KEY DEVELOPMENT WARNING SIGNS & DELAYED MILESTONE CAUSES

1. Key Development Warning Signs



Discrepant Head Size or Crossing Centile Lines
(too large or too small)





Persistence of Primitive Reflexes > 6 months





No Response to Environment or Parent by 12 months





Not Walking by 18 months





No Clear Spoken Words by 18 months





No Two-Word Sentences by 2 years





Problems with Social Interaction at 3 years





Congenital Anomalies, Odd Facies





Any Delay or Failure to Reach Normal Milestones





NOTE: PARENTAL CONCERNS MUST ALWAYS BE TAKEN SERIOUSLY.



4. Causes of Delayed Speech and Language

Definitions:

Delayed Speech:
No words till 3 years of age.



Delayed Language:
Defect in communication (guided by vision, hearing, and social contact).



Causes of Delayed Speech and Language:

* **Deafness**



* **Emotional and Social Deprivation**



* **Mental Retardation**



* **Paralysis in Muscles of Tongue and Articulation**



5. Causes of Delayed Walking

Definition:

Delayed Walking:
No walking by age 18 months.



Causes of Delayed Walking:

* **Cerebral Palsy**



* **Mental Retardation**



* **Peripheral Nerves Disorders**



* **Muscles Disorders**



* **Chronic Systemic Disorders**



CHAPTER 02

II. Nutrition



ADVANTAGES OF BREAST FEEDING & CONTRAINDICATION • NUTRITIONAL DISORDERS: KWASHIORKOR & MARASMUS • NUTRITIONAL DISORDERS: RICKETS

ADVANTAGES OF BREASTFEEDING

A. TO THE INFANT (Advantages)

1. Ideal Biochemical Composition:

- * **High content of protein:** highly nutritive. 
- * High content of **IgA, macrophages, PNL, and bifidus factor** → highly protective. 
- * **Less CHO** → less abdominal distension. 
- * **Low fat content** → easily digested. 
- * Contains **mucosal maturation factor** (epidermal growth factor) → **brush border formation, stimulates enzymes formation, intestinal motility** → passage of meconium → protects against jaundice. 

- 2. **Convenience:** Available anytime, anywhere, always fresh and sterile with proper temperature. 

- 3. **Anti-allergic:** Less incidence of **bronchial asthma, eczema.** 

- 4. **Cheaper.** 

- 5. **Psychological Advantages:** Establishing the **maternal-infant bond.** 

- 6. **Anti-infective Properties:** Breast milk contains:

- * **Immunoglobulins**, especially **secretory IgA** against different pathogens (e.g., E. coli, Shigella, Salmonella, viruses like poliovirus, etc.). 
- * **Phagocytes & lymphocytes.** 
- * **Lactoperoxidase**, which protects against different bacterial pathogens. 
- * **Lysozymes** that destroy bacterial cell walls. 
- * **Bifidus factor:** Stimulates the growth of *Lactobacillus bifidus*, which produces lactic acid, changing intestinal pH → **unsuitable medium for pathogenic organisms**; also important for **Vit. K production.** 
- * **Lactoferrin:** An iron-binding protein that **prevents iron uptake by organisms**, causing a bacteriostatic effect on E. coli. 
- * **Complement:** Important for **opsonization** of organisms. 
- * **Interferon:** Decreases intracellular **viral multiplication.** 
- * **B₁₂ binding factor** (B₁₂ is essential for many bacteria).

B. TO THE MOTHER (Advantages)



- 1. **Helps involution** of the birth canal.



- 2. **Natural method of contraception.**



- 3. **Decreases the incidence of breast cancer.**

CONTRAINDICATIONS OF BREASTFEEDING

1. Maternal Contraindications:

- * **Relative / Temporary Contraindications** (requires regular evacuation of the breast):



- * **Acute mastitis** and **breast abscess** until recovery.



- * **Chronic illness** till controlled.



- * **Maternal medication** that can pass in milk until stopped.



- * **Maternal Hepatitis B**, unless the newborn receives Hepatitis B immunoglobulin and Hepatitis B vaccine.

- * **Absolute Contraindications** (use bromocriptine to stop milk secretion):



- * **Cancer breast.**

- * **Insanity of the mother.**



- * **HIV infection.**

2. Infant Contraindications:

- * **Relative Contraindications:**



- * **Acute illness** until recovery (e.g., acute heart failure, pneumonia).



- * **Breast milk jaundice:** stop breastfeeding for 48-72 hours.



- * **Congenital anomalies** until surgical correction.

- * **Absolute Contraindications:**



- * Some **inborn errors of metabolism**, e.g., galactosemia.

UNDERSTANDING PEM: KWASHIORKOR vs. MARASMUS

Protein-Energy Malnutrition: Definitions, Etiology, Clinical Features, and Key Differences.

KWASHIORKOR

Definition

A clinical condition resulting from **deficiency of protein** in the diet in the presence of **almost normal caloric intake**, usually occurring in the 2nd or 3rd years of life.



Etiology

- Primary (Dietetic): **Unbalanced diet, faulty weaning** (abnormal quality, abnormal time).
- Secondary: Anorexia, stomatitis, malabsorption, gastroenteritis, infections.



Clinical Picture - Constant Features (All must be present, or pre-kwashiorkor):

1. **Growth failure:** Failure to gain proper weight (Weight for age < 80%); delayed bone age.



2. **Pitting oedema:** Starts in feet, puffy face with prominent cheeks (moon face); ascites absent. (Causes: decreased plasma proteins, increased plasma aldosterone/ADH, oxidative stress).



3. **Mental changes:** Apathy, no interest in surroundings, **no smile**. (Caused by deficient amino acids and trace elements Cu, Mg, Zn).



4. **Muscle wasting with preserved subcutaneous fat.**



Clinical Picture - Non-Constant / Variable Features:



1. **Skin changes:** Erythema, hyperpigmentation, desquamation → ulceration, fissuring, crackling. Often at pressure/flexural sites. (Deficiency of essential fatty acids, amino acids, Vit A, Zn).



2. **Hair changes:** Loses luster, dry, sparse, easily pickable. Progressive lightening (black-brown → red → orange → yellow). **Flag Sign** (alternating bands). (Deficiencies in essential amino acids, Vit A, Zn).



3. **Anemia:** Due to iron (microcytic), Vit B12/folic acid (megaloblastic), or infections (normocytic), or hemolysis.



4. **Associated Deficiencies:** Vit A (xerophthalmia, Bitot spots), Vit B2 (angular stomatitis, glossitis), Vit B12/Folic Acid, Vit D (atrophic rickets), Vit C (Scurvy).



5. **GIT changes:** **Hepatomegaly** (fatty infiltration), **Diarrhea** (malabsorption/infections), **Abdominal distension** (hypokalemia/toxic ileus).



• **Causes of Death:** Recurrent infections, electrolyte imbalance (refeeding/gastroenteritis), hypothermia, hypoglycemia, heart failure.



MARASMUS

Definition

A chronic state of PEM resulting from a **low intake of energy**, commonly following repeated **infection**; most cases are in the first 2 years of life.



Etiology

- Primary (Dietetic/Nutritional): **Early failure of breastfeeding, delayed weaning.**
- Secondary (Non-Dietary): **Defective intake** (chronic vomiting, anorexia), **impaired absorption** (malabsorption), **increased nutrient loss** (persistent diarrhea), **increased requirements** (CHD).



Clinical Picture - Constant Features:

1. **Weight for age < 60%** of the reference with **no oedema**.



2. **Loss of subcutaneous fat:** First abdomen, then limbs, finally buccal pad of fat (→ senile face).



3. **Muscle wasting:** Evident by **decreased mid-arm circumference** less than 12.5 cm in children 1-5 years.



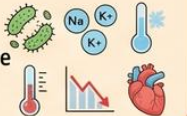
Clinical Picture - Variable Features:

• **Anxious, irritable, and hungry** (unless listless with infection); associated deficiencies similar to kwashiorkor.



Causes of Death:

Match kwashiorkor (Recurrent infections, electrolyte imbalance, hypothermia, hypoglycemia, heart failure).



KEY DIFFERENCES BETWEEN MARASMUS & KWASHIORKOR

Usual Age		Muscle Wasting & Growth Retardation		Skin Changes	
Marasmus 0-3 years	vs	Kwashiorkor 1-3 years		Seldom	Often diffuse depigmentation, flaky paint/ais
Edema		Mental Changes		Hair Changes	
None	vs	Lower legs, face, or generalized		Seldom	Often sparse, straight, silky, depigmentation, gray/red
Wasting		Appetite		Serum Albumin & Urinary Urea	
Gross loss of subcutaneous fat; skin and bone	vs	Sometimes hidden, fat blubbery appearance		Usually good	Usually poor
				Usually normal or low	vs Low

INFANTILE NUTRITIONAL & REFRACTORY RICKETS: A COMPREHENSIVE GUIDE

INFANTILE NUTRITIONAL RICKETS



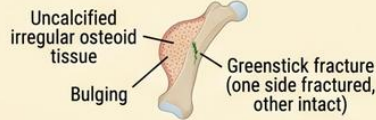
DEFINITION: "Systemic disease; failure of mineralization in growing bone."

PATHOLOGY OF RICKETS

DEFINITION

ACTIVE RICKETS

Cartilage cells fail to degenerate, continue proliferation. Accumulation of non-rigid osteoid. ZPC disappears.



HEALING RICKETS

Calcium deposition in ZPC (epiphyseal line).



HEALED RICKETS

Mineralization of ZPC completed. Normal bone density.



BIOCHEMICAL CHANGES IN ACTIVE RICKETS:



Vit D Deficiency: → Less Ca absorption from intestine/kidney
→ Slight lowering of serum Ca. PTH increases to maintain normal Ca (mobilizes from bone).



Serum Phosphorus: Decreased (secondary hyperparathyroidism).

Serum Alkaline Phosphatase: Increased (increased osteoblastic activity).

SUMMARY OF VALUES:

Serum Ca: Normal (9-11 mg%); Serum P: Low (4.5-6.5 mg%);
Serum Alk Phos: Increased (first to rise, last to normalize).

NOTE ON HYPOCALCEMIA & TETANY:

May occur if parathyroid fails, bone stores exhausted, Vit D shock without Ca, or hyperventilation → **alkalosis tetany**.

CLINICAL PICTURE:

EARLY CASES (EARLY DETECTION):



Disturbed ANS: Irritability, anorexia, excessive frontal sweating.



Craniotabes: Abnormal softness (squash ball sensation) in occipital bones, disappears by 1st year.



Rachitic Rosary: Palpable enlargement of costochondral junction.



Broadening: Thickening of wrists & ankles.



High serum Alkaline Phosphatase.



MUSCULAR HYPOTONIA & LAXITY: Due to hypophosphatemia; delayed milestones, viscerotoposis, spinal issues.

ADVANCED RICKETS:

SKELETAL MANIFESTATIONS:

Head: Frontal/parietal bossing (box-shaped skull), wide fontanelles, delayed teeth eruption, caries.

Thorax: Rachitic rosaries (visible), Harrison's sulcus (groove, flaring below), Pigeon chest (protruded sternum).

Spinal Column: Kyphosis, scoliosis, lumbar lordosis.

Limbs: Broadening of wrists/ankles, Marfan's sign (transverse groove on malleoli), Genu varum (Bow legs), Genu valgum (Knock knees), Greenstick fractures.

NEUROLOGICAL: Hypotonia, **INCREASED INFECTION SUSCEPTIBILITY:** Recurrent chest infections, T&B lymphocytes dysfunction.

INVESTIGATIONS:



Biochemical: Serum Ca Normal, P, Serum Alk Phos Increased, 1,25 dihydroxy vit D Decreased



Radiological (Wrist X-ray):

- Epiphyseal changes:** Broadening, cupping, fraying of lower radius & ulna.
- Diaphyseal changes:** Rarefaction, greenstick fractures, deformities.
- Increased radiological joint space.

PREVENTION & TREATMENT



PREVENTION: UV exposure, daily Vit D (400 IU term, 1000 IU premature), proper weaning.



TREATMENT: Oral Vit D (1500-5000 IU until recovery), or Shock therapy (600,000 IU).
Adjuvant therapy: Adequate Ca, P, UVR. Treat complications (infections, tetany).

REFRACTORY RICKETS

DEFINITION: "Rickets resistant to usually recommended doses of Vitamin D."

A. PRIMARY HYPOPHOSPHATEMIC RICKETS



Genetics: X-linked dominant, defect in phosphate reabsorption & Vit D conversion.



Clinical: Onset 2nd year, late rickets, short stature, marked deformities, good muscle tone.



Investigations: Serum Ca Normal, P Low, Alk Phos Increased, 1,25 Vit D Decreased, Urinary Phosphate Increased



Treatment: Oral phosphate, calcitriol

B. VITAMIN D DEPENDENT RICKETS



Type I: Autosomal recessive, renal 1α hydroxylase defect. Severe hypocalcemia. Responds to 1,25 dihydroxy Vit D.



Type II: Autosomal recessive, target organ resistance. Requires very high doses of calcitriol.

D. FANCONI SYNDROME



Definition: Multiple renal tubular defects: glycosuria, aminoaciduria, phosphaturia, bicarbonate loss → metabolic acidosis.



Manifestations: First 2 years: rickets, polyuria, polydipsia, dehydration, failure to thrive.



Treatment: Large doses Vit D, phosphate, acidosis correction.

E. LOWE SYNDROME (OCULO-CEREBRORENAL DISEASE)



Genetics: X-linked recessive.



Manifestations: Fanconi syndrome + congenital cataract/glaucoma + mental retardation/hypotonia.

Investigations & Treatment: Same as Fanconi syndrome.



Genetics: Autosomal recessive, marked deficiency of alkaline phosphatase.

Clinical Types: Congenital lethal, Hypophosphatasia tarda (rickets, deformities)

Treatment: No effective treatment

C. RENAL OSTEODYSTROPHY (UREMIC RICKETS)



Definition: Skeletal changes in chronic renal failure.



Mechanism: Renal hydroxylation failure, filtration defect → high P → secondary hyperparathyroidism → osteitis fibrosa cystica.



Manifestations: Rickets deformities, hyperphosphatemia, hypocalcemia, raised Alk Phos, metabolic acidosis, high urea & creatinine.



Treatment: Renal dysfunction treatment, calcitriol, high Ca, low P, phosphate binders.

F. HYPOPHOSPHATASIA



Genetics: Autosomal recessive, marked deficiency of alkaline phosphatase.



Clinical Types: Congenital lethal, Hypophosphatasia tarda (rickets, deformities).

Treatment: No effective treatment.

DIAGNOSTIC APPROACH TO REFRACTORY RICKETS:

- 1st Step: Measure Serum Phosphorus**
- High P → Chronic Renal Failure (Renal Osteodystrophy).
 - Low or Normal P → Proceed to 2nd Step.

- 2nd Step: Measure Blood pH**
- Low pH (Acidosis) → Fanconi Syndrome.
 - Normal pH → Proceed to 3rd Step.

- 3rd Step: Measure Serum Calcium**
- Low Ca → Vit D Dependent Rickets Type I and Type II.
 - Normal Ca → Primary Hypophosphatemic Rickets.

CHAPTER 03

III. GIT



COW'S MILK ALLERGY IN PEDIATRICS • GERD & HYPERTROPHIC PYLORIC STENOSIS • VOMITING IN PEDIATRICS • ABDOMINAL PAIN IN PEDIATRICS

COW'S MILK ALLERGY (CMA): Comprehensive Overview

DEFINITION & EPIDEMIOLOGY

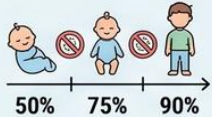
Cow's Milk Allergy (CMA) is an immunologically mediated adverse reaction to proteins in cow's milk, mainly in infants and young children

It is the **most common food allergy** in infancy

Prevalence: **2-3% in infants** and **less than 1% in adults**

Usually presents in the **first year of life**

Majority develop **tolerance over time**:
50% by 1 year
75% by 3 years
90% by 5 years



CLASSIFICATION OF CMA

1. IgE-mediated CMA: **Immediate reactions** (minutes to 2 hours), mediated by **IgE antibodies**



2. Non-IgE-mediated CMA: **Delayed reactions** (hours to days), driven by cell-mediated response



3. Mixed CMA: Clinical features of **both** IgE and non-IgE mechanisms

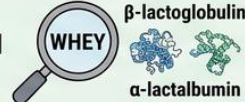


COW'S MILK PROTEINS

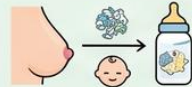
Major allergenic proteins: **Casein (80%)** and **Whey proteins (20%)**



Whey consists of **β-lactoglobulin** and **α-lactalbumin**



Proteins pass through **breast milk** or **formula feeding**



CLINICAL MANIFESTATIONS

IgE-Mediated CMA (Acute Signs)

Skin: Urticaria, angioedema, flushing, pruritus



GI: Vomiting, abdominal pain, diarrhea



Respiratory: Wheeze, cough, stridor, rhinitis



Systemic: Life-threatening anaphylaxis



Non-IgE-Mediated CMA (Delayed Signs)

GI (Most Common): Chronic diarrhea, blood/mucus in stool, vomiting, abdominal distension, constipation, failure to thrive



Skin: Chronic, refractory atopic dermatitis



CLASSIFICATION OF CMA

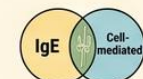
1. IgE-mediated CMA: **Immediate reactions** (minutes to 2 hours), mediated by **IgE antibodies**



2. Non-IgE-mediated CMA: **Delayed reactions** (hours to days), driven by **cell-mediated response**



3. Mixed CMA: Clinical features of **both** IgE and non-IgE mechanisms



CMA vs. LACTOSE INTOLERANCE COMPARISON

Clinical History (Cornerstone): Type of feeding, timing, nature of symptoms, family history



Skin Prick Test (SPT): Identifies IgE-mediated CMA; positive points to allergy



Serum Specific IgE: Supports IgE-mediated CMA, cannot confirm alone



Elimination Diet: Complete removal for **2-4 weeks**; clinical improvement supports diagnosis



Oral Food Challenge (Gold Standard): Strict medical supervision; contraindicated in severe anaphylaxis



MANAGEMENT PROTOCOL

General Principles: Complete **elimination**, adequate nutrition, regular follow-ups



Breastfed Infants: Continue breastfeeding; mother **strictly avoids dairy**; mother receives calcium & Vitamin D



Formula-Fed Infants: **Extensively Hydrolyzed Formula (eHF)** first-line; avoid standard/partial/goat; **Amino Acid Formula (AAF)** for severe cases



Complementary Feeding: Avoid all dairy; read labels for **hidden milk sources** (biscuits, processed foods)



Acute Reactions Treatment: Antihistamines for mild/moderate; **Intramuscular epinephrine**, urgent referral, parent education for **anaphylaxis**



PROGNOSIS & PREVENTION

Prognosis: Regular reassessments (6-12 months), food challenges, **excellent overall prognosis** in majority



Prevention: **Exclusive breastfeeding** highly recommended; no role for delaying complementary feeding; milk avoidance during pregnancy not supported (unless mother is allergic)



[Image Missing: GERD & HPS.jpeg]

NAUSEA & VOMITING: DEFINITIONS, PATHWAYS, DIAGNOSIS & MANAGEMENT

An Explanatory Graphic

1 CORE DEFINITIONS

Nausea: An unpleasant sensation of the **imminent need to vomit**, usually referred to the **throat** or **epigastrium**; a sensation that may or may not ultimately lead to the act of vomiting.



Vomiting: Forceful oral expulsion of gastric contents associated with the contraction of the **abdominal and chest wall musculature**.



Regurgitation: The act by which food is brought back into the mouth without the abdominal and diaphragmatic muscular activity that characterizes true vomiting.



2 GENERAL PATHWAYS & DIFFERENTIAL DIAGNOSIS BY AGE

- Pathological vomiting** stems from four primary diagnostic categories: **Gastrointestinal (GIT), Infections, Metabolic disorders, or Increased Intracranial Pressure (ICP).**



Infant Etiologies

- Gastro-oesophageal reflux, feeding problems, acute infections** (gastroenteritis, respiratory tract infections/otitis media, whooping cough/pertussis, urinary tract infections, meningitis), **dietary protein intolerances, metabolic disorders, congenital adrenal hyperplasia, renal failure, and mechanical intestinal obstructions** (pyloric stenosis, duodenal or other site atresias, intussusception, malrotation, volvulus, duplication cysts, strangulated inguinal hernia, or Hirschsprung disease).



Preschool Children Etiologies

- Gastroenteritis, systemic infections** (respiratory tract/otitis media, urinary tract, meningitis, whooping cough), **acute appendicitis, raised intracranial pressure, coeliac disease, renal failure, inborn errors of metabolism, torsion of the testis, and mechanical intestinal obstructions** (including intussusception, malrotation, volvulus, post-surgical adhesions, or foreign body-bezoar).



School-Age and Adolescent Etiologies

- Gastroenteritis, severe systemic infections** (including pyelonephritis, septicaemia, and meningitis), **peptic ulceration** alongside H. pylori infection, **acute appendicitis, migraine headaches, raised intracranial pressure, coeliac disease, renal failure, diabetic ketoacidosis, alcohol/drug ingestion or specific medications, cyclical vomiting syndrome, bulimia/anorexia nervosa, pregnancy, and acute torsion of the testis.**



3 DIAGNOSTIC WORKUP & MANAGEMENT

Clinical evaluation always begins with a detailed **history** and **physical examination**, which is often completely sufficient on its own without requiring additional testing.



A **diagnostic workup** can involve **blood tests, urine tests, or abdominal/chest X-rays** depending entirely on which underlying clinical condition is suspected.



The core strategy of **management** centers on keeping children thoroughly **hydrated** and initiating **targeted treatment** for the underlying cause.



Introduction & Classification of Pediatric Abdominal Pain.



Abdominal pain is one of the **most common clinical complaints** in childhood.

Challenge: Distinguishing self-limited conditions from **life-threatening conditions**



Two Main Clinical Paths:

1. **Acute Abdominal Pain**
2. **Chronic / Recurrent Abdominal Pain (RAP)**



Acute Abdominal Pain: Characteristics & Causes



Duration < 4-6 weeks
(sub-acute < 12 weeks)



Single, isolated episode of sharp/stabbing, **localized** pain

Acute Abdominal Pain Causes



Surgical Causes: Acute appendicitis, Intussusception, Obstruction, Renal calculus, Torsion, Trauma, Peritonitis, Meckel's



GIT: Gastroenteritis, IBD, Lymphadenitis, Constipation, Food poisoning, Ulcer, Lactose intolerance



Hepatobiliary & Splenic: Hepatitis, Pancreatitis, Cholecystitis, Splenic infarction/rupture



Genitourinary: UTI, Calculi, Threatened abortion, PID, etc.



Metabolic: DKA, Hypoglycemia, Porphyria, Adrenal insufficiency



Hematological: Sickle cell, Henoch-Schönlein, HUS



Toxins, Pulmonary & Misc: Lead poisoning, Pneumonia, Colic, Functional, etc.

The Acute Abdomen Presentation



Sudden onset, severe pain, developing rapidly



Key physical signs:
Guarding (reflex contraction)
Rigidity (involuntary hardness)
Rebound tenderness (pain on sudden release)



Recurrent Abdominal Pain (RAP) & Criteria



Most common pediatric pain (<15 yrs). Underlying cause often **undefined**.

Modified Apley and Naish Criteria:



3+ episodes over 3 months



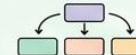
Severely affects **daily activities**, with **symptom-free intervals**



Severe phase lasts **> 3 minutes**



Peak incidence **5-7 years**; declines around **10 years**



Classification: Dysfunctional, Organic, or Psychiatric-related

Dysfunctional Abdominal Pain.



Complete absence of structural abnormality, inflammation, or injury.



Rome IV criteria assist diagnosis (e.g., IBS, functional dyspepsia, migraine).



Clinical criteria: Paroxysmal, variable severity, periumbilical, poor food relation



Associated autonomic symptoms (~30%): pallor, nausea, fatigue, anxiety

Organic Pain Diagnostic Red Flags.



1. **Clearly localized** (away from umbilicus)



2. **Actively radiates**



3. **Wakes child** from sleep



4. **Persistent fever**



5. **Involuntary weight loss**



6. **Tenderness or organomegaly**



7. **Blood in stools**



8. **Altered bowel movements**



9. **Lab evidence of anemia**



10. **Urinary symptoms**



11. **Elevated inflammatory markers (ESR, CRP)**



12. **Systemic features** (rash, joint pain).



CHAPTER 04

IV. Genetic Diseases



CHROMOSOMAL DISORDERS: DOWN SYNDROME • CHROMOSOMAL DISORDERS: TURNER SYNDROME • PRENATAL DIAGNOSIS (IMPORTANCE, INDICATIONS & TYPES)

UNDERSTANDING DOWN SYNDROME

Incidence & Occurrence



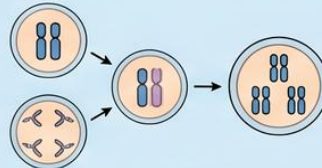
Incidence: Estimated in **0.7%** of live births, in **≈ 2%** of all pregnancies in women **over 35yrs**, and in **50%** of all spontaneous **1st trimester abortion**.



Occurrence: **More with advanced maternal age: 1 / 20 by age 45 years** and it **increases the risk of spontaneous abortion**. Males affection equals females.

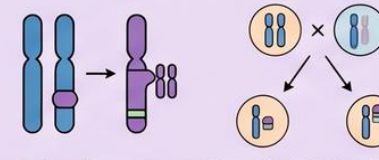
Cytogenetic Types (Mechanism & Risk)

Nondisjunction Type (94% - Most Common)



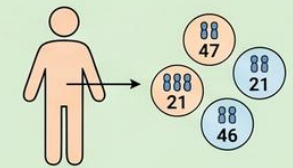
Extra chromosome 21 in **every cell**. **Risk increases with advanced maternal age**. Not familial with **low recurrence (<1-2%)**.

Translocation Type (3-4% - Familial)



Extra chromosome 21 material **attached (translocated)** onto another chromosome. Called **familial type with increased recurrence risk (2-3%** if translocated father or 10-15% if translocated mother). Parent may be a carrier of a **balanced translocation**.

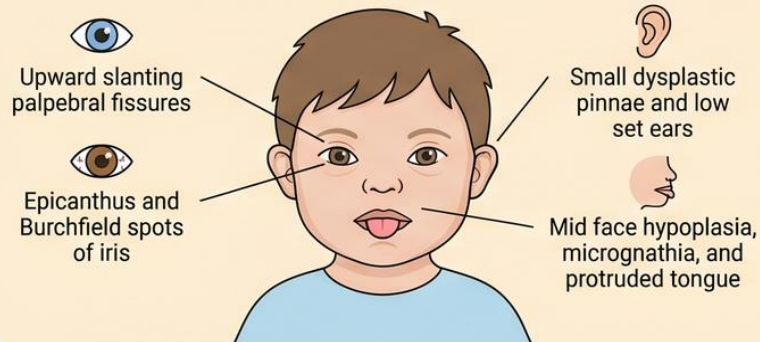
Mosaic Type (1-2% - Less Severe)



Only **some of the cells** have an extra chromosome 21, others are normal. Not familial with **low recurrence risk** and usually **less severe features**.

Clinical Phenotype

Dysmorphic Features



Body System Affection



Cardiac: Congenital heart defects (50%), particularly **septal defects (VSD)**.



Gastrointestinal: Duodenal/esophageal/anal atresia, TE fistula, **Hirschsprung disease**, chronic constipation and poor feeding, vomiting, aspiration.



CNS: Microcephaly, mental retardation, generalized hypotonia and premature Alzheimer.



Orthopedics: Short fingers, curved 5th finger, transverse palmer crease, wide gap between 1st and 2nd toes, **lax joints** including **dysplastic hips**, vertebral anomalies including atlantoaxial instability.



Endocrine: Hypothyroidism, obesity, and insulin resistance.



Blood and skin disorders: 1% lifetime risk of **leukemia**.

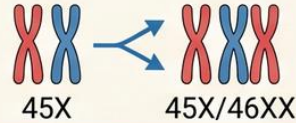


Other problems: Visual and hearing difficulties, **dental problems**, speech problems, and sleep apnea.

TURNER SYNDROME: Genotype & Clinical Overview

GENOTYPE

45X (most common) / Mosaic (45X/46XX)



CLINICAL PHENOTYPE: Life Stage Presentation

INTRAUTERINE

May be presented with **polyhydramnios** and **lung hypoplasia**.

Polyhydramnios



Lung Hypoplasia

NEONATAL

May be presented with **lymphedema of hands and feet**, **low posterior hair line** and **cystic hygroma**.



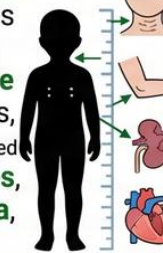
Lymphedema

Low Hair Line

Cystic Hygroma

CHILDHOOD

Intelligence usually **normal** (mild learning disabilities possible), **short stature**, **short webbed neck**, **wide carrying angle** at elbows, **broad chest** (widely spaced nipples), **renal anomalies**, **coarctation of the aorta**, **bicuspid aortic valve**.



ADOLESCENCE

Gonadal dysgenesis lead to **infertility**, **primary amenorrhea**, lack of development of **secondary sexual characteristics**: absence of breast development and failure to menstruate and increased risk of **hypertension**.



No Period



Hypertension Risk

PROGNOSIS

Normal life expectancy (if no complications) / **Infertile**.



Infertile

MANAGEMENT

Early screening for **cardiac disease**, **growth hormone therapy** for short stature and **estrogen replacement** at time of puberty.



Cardiac Screening



Growth Hormone Therapy



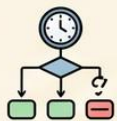
Estrogen Replacement

PRENATAL DIAGNOSIS: IMPORTANCE & METHODS

IMPORTANCE OF PRENATAL DIAGNOSIS



To **detect** changes in a fetus's **genes or chromosomes** before birth.



To **manage** the remaining weeks, **determine outcome**, **decide** whether to continue the pregnancy.



To **plan for possible complications** with the birth process or problems that may occur in the newborn infant.



To find conditions that may **affect future pregnancies**.

INDICATIONS: POPULATION AT RISK



Those at increased risk of genetic disorders as:



Women aged **35 or more** or have **recurrent abortions**



Previous child with **chromosomal abnormalities** or **multiple congenital anomalies**



Carriers for X-linked inherited disorders such as Duchenne muscle dystrophy and Fragile-X syndrome or **Parents who are carriers for chromosomal abnormalities**



Inborn errors of metabolism



Neural tube defects:
Anencephaly – encephalocele – spina bifida

TYPES OF PRENATAL SCREENING TESTS



NON-INVASIVE SCREENING TESTS



A. Biochemical markers:

Is routinely used to investigate women at risk of having a **Down syndrome** or other genetic conditions.

Triple tests result in some genetic disorders:

	uES	HOG
Open neural tube defect:	AFP Increased	Not applicable
Down syndrome:	AFP Decreased	Increased
Turner syndrome:	AFP Normal/Decreased	Increased
Edward syndrome:	AFP Decreased	Decreased

Some prenatal tests using biochemical markers:

- Triple test: hCG, AFP, uE3
- Quadruple test: hCG, AFP, uE3, inhibin A
- Combined test: NT, hCG, PAPP-A
- Integrated test: NT, PAPP-A + hCG, AFP, uE3, inhibin A



B. Isolation of Fetal cells from maternal circulation

- Isolation of **fetal nucleated cells** enabling investigation of **fetal chromosomes** in early pregnancy (FISH for numerical aberrations only)
- Isolation of **fetal DNA** to screen paternally inherited single-gene disorders or mutated genes.
- It can screen for **Duchenne muscle dystrophy**, **fetal RhD blood type**, **sickle cell anemia** and **beta-thalassemia** in heterozygous carriers.



C. Ultrasound

- Fetal abnormalities in the first trimester including changes in the head circumferences; heart, nuchal fold thickness, fetal and limbs size are indicators of chromosomal anomalies and several disorders.
- The second trimester ultrasound findings assess the fetal measurements and can assess fetal age.



INVASIVE SCREENING TESTS



A. Chorionic villus sampling:

- Can be done at about **10-11 weeks**, either transvaginal or transabdominally
- There is a small chance of **maternal cell contamination** and a **1-4% risk of miscarriage**, infection, placental trauma, or **feto-maternal hemorrhage**.



B. Amniocentesis:

- Can be done at **16-18/20 weeks**, transabdominally
- Miscarriage**, intrauterine infection, persistent amniotic fluid leak, **feto-maternal hemorrhage**, fetal injuries are less frequent complications



C. Fetal blood sampling:

- Blood** is taken from the umbilical vein at the placental insertion. **Fetal loss rate is variable** from 0-24%-12%
- Rapid assessment** of fetal chromosomal anomalies, haemoglobinopathies, coagulopathies, infection and immunodeficiency later in pregnancy



D. Fetoscopy

- Permit **direct access to fetus** via percutaneous introduction of small fiberoptic telescope in amniotic cavity
- Blood from fetus** to diagnose blood and metabolic diseases. Also, **fetal malformation visualization**



E. Fetal tissue sampling

- Fetal biopsy** for histological and ultrastructure studies as for prenatal diagnosis of **epidermolysis bullosa lethalis**. Fetal loss is as high as 1%.

CHAPTER 05

V. Endocrinology



SHORT STATURE • HYPOTHYROIDISM • DIABETES MELLITUS TYPE 1 & DKA • CHILDHOOD OBESITY

SHORT STATURE: DEFINITION, VARIANTS, PATHOLOGY, DIAGNOSIS & TREATMENT

1. SHORT STATURE: DEFINITION & NORMAL VARIANTS

DEFINITION

Height: ≤ 2 Standard Deviations (SD) below the mean for sex and chronologic age.

Growth Velocity: Below the 25th percentile over 6–12 months of observation.

Genetic Context: Short for the Mid-Parental Height (MPH).



MID-PARENTAL HEIGHT (MPH) FORMULAE

Girls: $MPH = (\text{Mother's Ht} + \text{Father's Ht} - 13) / 2$

Boys: $MPH = (\text{Mother's Ht} + \text{Father's Ht} + 13) / 2$

Note: Predicted adult height should be within ± 5 cm of the calculated MPH.



NORMAL VARIANTS OF GROWTH

FAMILIAL SHORT STATURE

- Genetic potential is low (fits the MPH).
- Growth Velocity:** Low-normal throughout life.
- Bone Age = Chronologic Age (Crucial differential point).



CONSTITUTIONAL DELAY OF GROWTH AND PUBERTY (CDGP)

- Normal birth size $\rightarrow$ downward shift at 3–6 months $\rightarrow$ grows parallel to but below the 3rd percentile by 3–4 years.
- Key features:** Delayed pubertal maturation + Delayed Bone Age (Bone Age < Chronologic Age).
- Outcome:** Catch-up growth occurs during delayed puberty, leading to relatively normal adult height.

IDIOPATHIC SHORT STATURE (ISS)

- Height ≤ 2 SD below the mean with no endocrine, metabolic, or systemic diagnosis.
- Normal/low-normal growth velocity and normal endocrine screening (including GH).



SMALL FOR GESTATIONAL AGE (SGA) INFANTS

Most achieve catch-up growth by 2 years of age to enter the normal range.



2. PATHOLOGICAL SHORT STATURE

TYPES OF SHORT STATURE

PROPORTIONATE SHORT STATURE (PSS)

- Body parts maintain normal proportions (e.g., Growth Hormone Deficiency).



DISPROPORTIONATE SHORT STATURE (DSS)

- Limbs are short relative to the trunk (e.g., Achondroplasia).



CAUSES OF SHORT STATURE

A. PRIMARY CAUSES

(Intrinsic to bone/genetics)

Skeletal Dysplasia: Achondroplasia, Osteogenesis imperfecta.

Chromosomal Abnormalities: Down syndrome, Turner's syndrome.

Intrauterine Growth Retardation (IUGR): Maternal infections, toxins, malnutrition.

Genetic Syndromes & Misc: Prader-Willi syndrome, Histiocytosis.



B. SECONDARY CAUSES (Systemic/Environmental)

Nutritional: Protein-energy malnutrition, Vitamin/mineral deficiency.

Gastrointestinal: Malabsorption (Celiac disease, Cystic Fibrosis), IBD.

Cardiovascular: Congenital Heart Disease (Large VSD, Tetralogy of Fallot).

Pulmonary: Asthma, Chronic suppurative lung disease.

Renal: Chronic uremia, Polycystic kidney, Pyelonephritis, Obstructive uropathy.

Endocrine: Hypothyroidism, Hypopituitarism, Diabetes Mellitus.

Immunological: Chronic infections (TB, AIDS), Juvenile Rheumatoid Arthritis.



3. EVALUATION & DIAGNOSIS

HISTORY

- Family history, birth weight, developmental milestones.

MEASUREMENTS

- Height, arm span, upper/lower segment ratio, head circumference plotted on growth charts.

PHYSICAL EXAM

- Check for dysmorphic features and systemic

RADIOLOGICAL

ECHOCARDIOGRAPHY

- To rule out Congenital Heart Diseases.

LABORATORY

- Basal and stimulated Growth Hormone (GH) levels, Thyroxine (T4), IGF-1, and sex hormones.

GENETIC ANALYSIS

- Karyotype. ****Mandatory Rule:** Karyotype is mandatory for any short female to exclude Turner syndrome.**

4. TREATMENT PROTOCOL

TREAT THE UNDERLYING CAUSE

- Surgery for pituitary tumors (e.g., Craniopharyngioma).
- Thyroid hormone replacement for Hypothyroidism.
- Nutritional rehabilitation for Malnutrition.

PANHYPOPITUITARISM REPLACEMENT

- Thyroid hormone, Hydrocortisone, and eventually sex hormones.

GROWTH HORMONE REPLACEMENT (rHuGH)

- Dose:** 0.18 – 0.3 mg/kg/week during childhood (higher doses during puberty).
- Route:** Subcutaneous (SC) injection, divided into 6 or 7 doses per week (daily).

CONGENITAL HYPOTHYROIDISM (CHT)

Deficient thyroid hormone action present at birth.

DEFINITION & OVERVIEW

- Deficient thyroid hormone action.
- Resulting from **absent/under-developed gland (DYSGENESIS)** or gland developed but cannot make hormone (**DYSHORMONOGENESIS**).
- The term '**congenital**' means present at birth.
- Common cause of preventable **intellectual disability**.
- Detected by **neonatal screening**. Early treatment prevents complications.

CAUSES OF CHT

Primary Hypothyroidism

- Defect of fetal thyroid development (**DYSGENESIS**): Aplasia, Hypoplasia, Ectopia.
- Defect in thyroid hormone synthesis (**DYSHORMONOGENESIS**).
- **TSH unresponsiveness & Resistance** to thyroid hormone.
- **Maternal antibodies**: TRBAbs.
- **Iodine deficiency** (endemic goiter).
- **Maternal medications**: Iodides, Amiodarone, PTU, Methimazole, Radioiodine.

Central Hypothyroidism

- Isolated **TSH deficiency**.
- Isolated **TRH deficiency**: mutation in TRH gene.
- **TRH unresponsiveness**: mutation in TRH receptor gene.
- Multiple congenital **pituitary hormone deficiencies**.

SYMPTOMS & PRESENTATION (Untreated)

- Most infants **asymptomatic** at birth. Full clinical picture by 3-6 mo.
- **Retardation of physical and mental development**.
- **Learning difficulties** or mild clumsiness.
- **Stunted growth**, short extremities, normal/increased head size.
- **Large anterior fontanel**, posterior fontanel may remain open.
- **Eyes far apart**, depressed broad nose bridge.
- **Narrow palpebral fissures**, swollen eyelids.
- **Mouth open**, thick protruding tongue.
- **Delayed dentition**.
- **Short thick neck**, fat deposits (clavicles/neck/shoulders).
- **Broad hands**, short fingers.
- **Dry and scaly skin**.
- **Myxoedema** (eyelids, hands, genitals).
- **Coarse**, brittle, scanty hair.

RADIOGRAPHIC FEATURES

- **Thyroid scan**: Missing or too small gland.
- **Plain radiograph**: Delayed skeletal maturation.
- **Bone age (Greulich & Pyle method)**: Absent/small epiphyses, stippled appearance, cortical thickening.

NEONATAL SCREENING

- All babies, **heel-prick blood test** (3rd-7th day).
- Tests for CHT by measuring **TSH**.
- **Diagnoses >95% of cases**: High TSH (>20 uIU/ml) & Low T4 (< 6.5 ug/dl).
- Confirmed by **venous blood sample**.

TREATMENT

- Treatment with **levothyroxine** (synthetic T4) started **without delay**.
- Dose **carefully calculated** based on **weight**, adjusted by TSH/T4 levels.
- Levothyroxine tablets can be **crushed in liquid**.

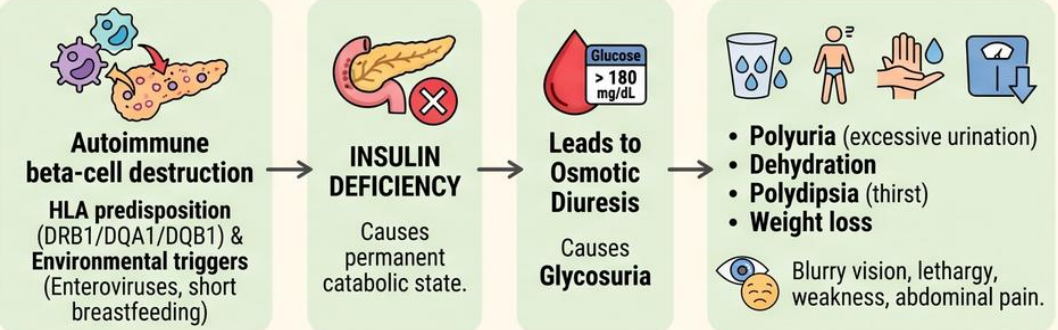
DIABETES MELLITUS (DM) & DIABETIC KETOACIDOSIS (DKA)

A Comprehensive Explanatory Overview

TYPE 1 vs TYPE 2 DM (Comparison)

TYPE 1	TYPE 2
 Sudden onset, children	 Gradual onset, adults
 Thin/normal body	 Often obese
 Ketoacidosis common	 Ketoacidosis rare
 Autoantibodies present (ICA, GAD-65, IAA, IA-2A)	 Autoantibodies absent
 Low/absent endogenous insulin	 Normal/increased insulin
 50% twin concordance	 90% twin concordance Genetic link stronger in T2

TYPE 1 DM PATHOGENESIS & PRESENTATION (Core Mechanism)



DIABETIC KETOACIDOSIS / DKA (Medical Emergency)



LIFE-THREATENING MEDICAL EMERGENCY with 10% MORTALITY
Precipitated by: Infection (UTI), Negligence/Omission, Trauma, Vomiting.

DIAGNOSIS & INSULIN THERAPY (Management & Criteria)

DIAGNOSIS CRITERIA



- HbA1C > 6.5%
- FPG > 126 mg/dL
- Random glucose > 200 mg/dL (with symptoms)

INSULIN THERAPY MANAGEMENT



Lines of Treatment: Diet, Exercise, Glucose Monitoring & Insulin



BOLUS (Meal):
Rapid/Short-acting (Lispro, Aspart) – 15 mins;
(Regular) – 30–60 mins before meals.



BASAL (Background):
Intermediate (NPH, Lente) – 10–16 hours;
Long-acting (Glargine, Detemir) – 20–24 hours.



Pre-mixed NOT recommended in pediatrics.

IMMUNE MARKERS & C-PEPTIDE



Autoantibodies:
GAD-65, ICA, IAA, IA-2A.

C-peptide < 0.2 nmol/l
confirms Type 1

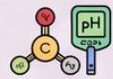
DIAGNOSTIC TRIAD



HYPERGLYCEMIA
(> 300 mg%)



KETONEMIA/KETONURIA



ACIDOSIS
(pH < 7.35, HCO₃ < 16 mEq/L)

CLINICAL PRESENTATION & SEVERITY



Signs: Kussmaul respiration (deep, rapid),
Acetone breath, Abdominal pain,
Dehydration (dry tongue, sunken eyes), Coma.

Mild

Mod

Severe

SEVERE: Venous pH < 7.15; HCO₃ < 10 mEq/L;
depressed respirations/coma.

MANAGEMENT PROTOCOL



- Immediate Hospitalization
- IV Fluids Correction
- Electrolyte Correction (esp. Potassium)
- Acid-base Balance
- Low-dose Continuous Insulin
- Treat Precipitating Causes.

DEFINITION & PERCENTILES

CHILDHOOD OBESITY

Obesity is a major pediatric health issue linked to childhood complications and lifelong risks.



Obesity/increased adiposity is defined using **BODY MASS INDEX (BMI)**, an excellent proxy for body fat.



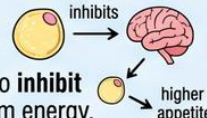
$$\text{BMI} = \frac{\text{weight in kg}}{\text{height in meters}^2}$$

Other measures: waist circumference (WC), hip circumference (HC), waist-to-hip ratio (WHR).

Pediatric Obesity Definition (Over 2 years, BMI charts)



Leptin Hormone Role: LEPTIN, released from fat cells, signals hypothalamus to **inhibit hunger** and regulate long-term energy. **Leptin levels connect directly to body fat amount.** Weight loss reduces leptin, potentially triggering appetite increases, making weight loss difficult.



COMPLICATIONS & DIAGNOSIS

Physical Complications



Type 2 Diabetes: Increased by obesity and sedentary lifestyle.



Cardiovascular Risk: High Cholesterol, High Blood Pressure, Metabolic Syndrome, dyslipidemia



Joint Pain: Hips, knees, and back pain/injuries due to extra weight



Breathing Problems: Asthma, Obstructive Sleep Apnea



NAFLD: Non-alcoholic Fatty Liver Disease, liver damage from fatty deposits

Social/Emotional Complications

Bullying, loss of self-esteem, depression, anxiety



Diagnosis & Evaluation



Clinical Assessment: Family history (nutritional/activity habits), pediatric exam for cause and comorbidities.



Laboratory Investigations: Transaminases (AST, ALT), basal glycaemia, lipid profile. **HOMA Index** (fasting glucose x insulin / 22.5) indicates **Insulin Resistance (IR)**, even with normal fasting glycaemia. Baseline screening for GH, IGF-I, IGFBP-3, iPTH, thyroxine, TSH, electrolytes (for hypercortisolism/pseudohypoparathyroidism).



OGTT: Oral-Glucose-Tolerance Test if family history of T2D or IR signs like Acanthosis nigricans/PCOS.



Imaging:

- Bone Age Radiograph** (Left hand/wrist): Shows **accelerated bone age** vs chronological age
- Abdominal Ultrasound:** If non-alcoholic steatohepatitis or PCOS is suspected

ETIOLOGY & RISK FACTORS

Complex Interaction: Obesity is a complex disease with multiple interacting factors increasing risk.

Environmental & Diet:

- Lifestyle:** Too little activity, too many calories.
- Diet:** High-calorie **fast foods**, baked goods, snacks, candy, **sugary drinks** (fruit juices, sports drinks) cause weight gain.



Lack of Exercise: Sedentary activities (TV, video games) burn fewer calories.



Family Factors: Family environment with available high-calorie foods and lack of encouraged physical activity.



Psychological: Stress (personal, parental, family) can lead to overeating for coping, emotions, or boredom.



Socioeconomic: Limited resources lead to convenience foods (frozen meals) and lack of safe exercise places.



Hormonal Factors: Underlying endocrinopathies like **Cushing syndrome** and **hypothyroidism**.



Genetic Factors: Syndromes such as **Prader-Willi Syndrome**.



Certain Medications: Prescription drugs like **prednisone**, **lithium**, **propranolol**.

DIFFERENTIAL DIAGNOSIS, PREVENTION & TREATMENT

Differential Diagnosis (High-Yield Contrast)



Nutritional Obesity: Consistent/accelerated growth, early puberty, advanced **bone age** (> 2 SD).



Endocrine Obesity: Decreased/decelerated linear growth. Test for thyroid hormones.



Genetic Syndrome Obesity: Severe obesity < 5 years. Developmental delay, short stature, dysmorphic facies, hyperphagia.



Screening: **DNA methylation**, exome sequencing, karyotyping for syndromes (Prader-Willi, etc.)

Prevention

Healthy family lifestyle (eating/activity), low-fat healthy snacks, expose to new foods, **non-food rewards**, adequate sleep



Treatment Lines

1. **Diet Control:** Appropriate calories (fruits, vegetables, whole grains, lean meat, fish, poultry). Avoid bad habits (eating in bedroom).



2. **Increasing Physical Activity:** Walking to school, leisure family activities, organized sports.



3. **Pharmacotherapy:** No effective reversal drug. FDA-approved: **ORLISTAT** for children < 16 years, decreases fat absorption, modest weight loss.



4. **Bariatric Surgery:** STRICT CRITERIA:

Complete/near-complete **skeletal maturity**, **BMI ≥ 40**, medical complication present, **FAILED ≥ 6 months** organized weight management



CHAPTER 06

VI. Hematology and oncology

IRON DEFICIENCY ANAEMIA • APLASTIC ANAEMIA • CHRONIC HEMOLYTIC ANAEMIA • RBC ABNORMALITIES: SPHEROCYTOSIS • RBC ABNORMALITIES: THALASSEMIAS • RBC ABNORMALITIES: G6PD DEFICIENCY • HEMOPHILIA • ITP (IMMUNE THROMBOCYTOPENIC PURPURA) • ALL & PROGNOSTIC FACTORS • HODGKIN LYMPHOMA

UNDERSTANDING IRON DEFICIENCY ANEMIA (IDA)

1. Definition & Prevalence



- Iron deficiency anemia (IDA) is the **most common type of anemia worldwide**.



- There are **low iron stores** leading to **decreased red blood cell production**.



- Iron is critical for **motor and cognitive development** in childhood, and for **physical activity**.

2. Causes of Iron Deficiency Anemia



Inadequate iron intake: Poor diet, **poverty**, ignorance, inadequate vitamin C.



Poor absorption: Due to less **Non-heme iron**, less **vitamin C**, or inhibitors like **phytates**, **oxalates**, **excess tea/coffee (tannin)**, **Eggs** (phosphate), and **Milk** (calcium). Also small intestine malabsorption disorders.



Increased iron demand: Periods of **rapid growth** like infancy, childhood and adolescence.



Excessive iron loss: **Menstruation**, intestinal worms, and **hookworm infestation**.

3. Clinical Manifestations



- Most common symptoms include **pallor**, **weakness**, and **fatigue**.



- Key signs are **glossitis**, **cheilitis** (inflammation/fissures of lips).



- and **pica**, which is the **intense craving for nonfood items** like clay, dirt, rocks, starch, chalk, soap, paper, or cardboard.

4. Diagnosis



Complete Blood Count (CBC) shows **microcytic hypochromic anemia**.



Iron profile shows **low serum iron**, **low serum ferritin**, **increased Total iron binding capacity (TIBC)**, and **decreased transferrin saturation (TS)**.

5. Prevention & Management



For **breastfed term infants**, introduce **iron supplementation between 4 to 6 months** of age, and at **2 weeks** of age for premature infants. Avoid **unmodified cow's milk** until age 1 year.



Iron replenishment includes **iron-rich foods** (liver, meat, poultry, fish) and **drug therapy** with **oral iron** (absorbed best in acidic environment) or **parenteral iron** (IM or IV if oral is ineffective).

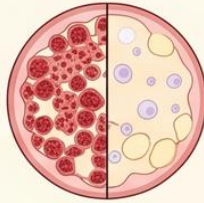


Blood transfusion is appropriate for very **severe IDA** where **hemoglobin concentration is below 4-5 g/dL** due to the risk of **cardiac failure**, **stroke**, and **death**.

APLASTIC ANEMIA: BONE MARROW FAILURE & PANCYTOPENIA

DEFINITION

Failure of the **bone marrow** to produce one or more blood cells, leading to a **reduction in Red Blood Cells (RBCs), White Blood Cells (WBCs), and Platelets.**



ETIOLOGY

CONGENITAL (Mostly Inherited):

Fanconi anemia and Diamond Blackfan syndrome.



ACQUIRED IDIOPATHIC:

Nearly **50%** of childhood-acquired aplastic anemia occurs sporadically for **unknown reasons.**



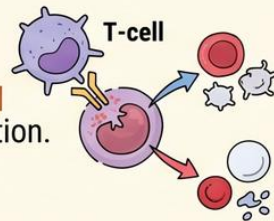
ACQUIRED SECONDARY:

Secondary to **virus infections** (hepatitis, EBV, CMV, parvovirus B19, HIV), **drugs** (chloramphenicol, anticonvulsants, chemotherapy), **toxins** (heavy metals, Benzene, environmental toxins, insecticides), exposure to **radiation**, or **autoimmune diseases** like SLE.



PATHOPHYSIOLOGY

Results from **autoimmune injury** to the **stem cell** causing **reduction** in one or all blood cell production.



CLINICAL MANIFESTATIONS



Pallor, fatigue (anemia), lack of energy, and easy **fatigue** due to anemia. **Recurrent infections** (leukopenia) like sinusitis, urinary tract infection, respiratory tract infection, and oral fungal infection due to leukopenia. **Easy bruising, purpura, and mucous membrane bleeding** due to thrombocytopenia. There is **characteristically no hepatosplenomegaly** and **no lymphadenopathy** on examination.



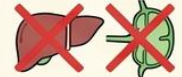
ANEMIA



INFECTION



BLEEDING



DIAGNOSIS & SEVERITY CLASSIFICATION



INVESTIGATIONS:

Complete blood picture reveals **pancytopenia** with **reticulocytopenia**. Bone marrow aspiration and biopsy show a great **reduction** in the number of cells, resulting in a **hypocellular marrow**.



SEVERE APLASTIC ANEMIA CRITERIA:

Hypocellular BM for age plus two of the following three criteria:

- Platelet count $< 20,000/\text{mm}^3$
 - Absolute reticulocyte count $< 40,000/\text{mm}^3$, or
 - Absolute neutrophil count (ANC) $< 500/\text{mm}^3$.
- Very severe** aplastic anemia is defined when the ANC count is $< 200/\text{mm}^3$.



TREATMENT

Stopping all offending drugs and removing causative agents. **Isolation** to prevent infection, and **restricted activity** to avoid trauma or bleeding.

Intravenous antibiotics as soon as fever appears.

Red cell transfusion usually when Hb is below 7 g/dL, and platelet transfusion when platelets are below 20,000/microliter.

Bone marrow transplantation (BMT) is indicated for severe aplastic anemia with a matched sibling donor.

Immunosuppressive therapy includes Antithymocyte globulin (ATG), cyclosporine, cyclophosphamide, and corticosteroids.



CHRONIC HEMOLYTIC ANEMIA & HEREDITARY SPHEROCYTOSIS: CLINICAL & PATHOPHYSIOLOGICAL OVERVIEW

1. The Classical Clinical Triad & Acute Crises

The Hemolytic Triad: Clinically Characterized

ANEMIA:



Presenting as progressive **skin & mucosal pallor**, **fatigue**, and **exercise intolerance**.

JAUNDICE:



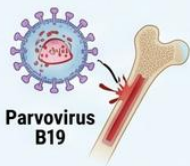
Manifesting as **scleral icterus**, accompanied by **dark-colored urine (bilirubinuria)**.

HEPATOSPLENOMEGALY (HSM):



Splenic enlargement driven by **chronic overwork** **clearing defective cells**, paired with variable liver enlargement.

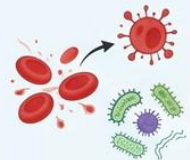
Life-Threatening Acute Crises: (Dangerous!)



1. APLASTIC CRISIS:

MOST DANGEROUS!

Almost universally triggered by **Parvovirus B19**. Completely **halts marrow erythropoiesis**. Sudden drop in hemoglobin & **profound reticulocytopenia (<1%)**.



2. HEMOLYTIC CRISIS:

Sudden, temporary **acceleration** of RBC destruction, precipitated by **acute bacterial or viral infections**.

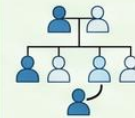


3. MEGALOBLASTIC CRISIS:

Due to **rapid, severe depletion** of **nutritional folate stores** driven by chronic, highly **accelerated bone marrow erythropoiesis**.

2. Hereditary Spherocytosis (HS): Pathophysiology

Definition & Genetics: Common Inherited Defect



MOST COMMON INHERITED RBC MEMBRANE DEFECT:

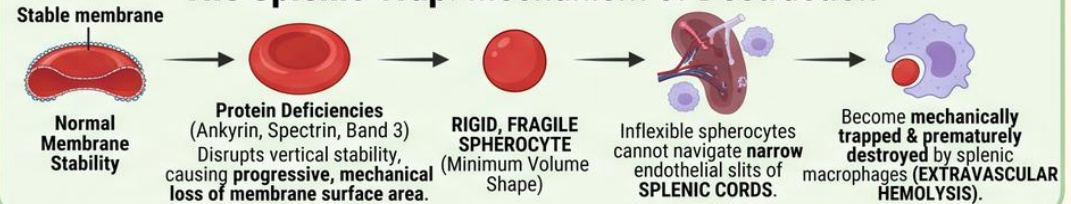
Primarily transmitted as an **AUTOSOMAL DOMINANT TRAIT**.



MOLECULAR TRAIT:

Caused by **genetic mutations** affecting vital structural **membrane proteins**: most frequently **ANKYRIN, SPECTRIN, or BAND 3**.

The Splenic Trap: Mechanism of Destruction



3. Clinical Presentation & Laboratory Diagnostics

Clinical Manifestations: Key Symptoms



NEONATAL JAUNDICE

Prominent, within the **first few days of life**.



MILD-TO-MODERATE CHRONIC ANEMIA

Ongoing condition.



PROGRESSIVE SPLENOMEGALY

Spleen enlargement over time.



HIGH RISK OF PIGMENT GALLSTONES

(Calcium bilirubinate cholelithiasis) in **early childhood** due to chronic hyperbilirubinemia.

Laboratory Diagnostics: Key Findings



COMPLETE BLOOD COUNT (CBC)

Normocytic/mild microcytic anemia. **Elevated MCHC (>36 g/dL)** (Key Signature: cellular dehydration). **Reticulocytosis (>2%)** (Bone marrow compensation).



OSMOTIC FRAGILITY TEST

Demonstrates **significantly increased RBC lysis** in hypotonic saline due to **reduced surface-to-volume ratio**.



PERIPHERAL BLOOD SMEAR (PBS)

Directly confirms numerous **SPHEROCYTES**: small, dark, hyperchromic round RBCs **lacking central pallor**.

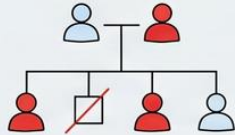


EMA BINDING TEST

(Flow Cytometry) **MODERN GOLD STANDARD**. Quantifies marked **reduction in fluorescent intensity bound to membrane proteins**. High sensitivity & specificity.

HEREDITARY SPHEROCYTOSIS (HS): AN EXPLANATORY OVERVIEW

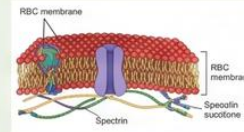
DEFINITION & GENETICS



Hereditary Spherocytosis (HS) is an inherited chronic hemolytic anemia. Mode of inheritance: **Autosomal Dominant (AD)** in **75%** of cases.



PATHOGENESIS



Dysfunction of **spectrin** or **ankyrin**, essential **cytoskeleton components**, responsible for RBC shape. Leads to a **loss of cell surface area**, resulting in a **spherical shape** and **premature destruction** in the **spleen**.



CLINICAL PICTURE



Presents with **anemia** and **hyperbilirubinemia** in the newborn period. **Splenomegaly** develops beyond infancy. Complications include **pigmentary bilirubin gallstones** and signs of chronic hemolytic anemia.

DIAGNOSIS

Complete Blood Count (CBC): Anemia (6-8 g/dL in severe cases), reticulocytosis, increased MCHC (>35 g/dL).

Peripheral blood smear: Spherocytes and microspherocytes.

Direct Antiglobulin Test (Coombs test): Negative. Increased osmotic fragility, and **Eosin-5-maleimide (EMA)** is the diagnostic test of choice.



COMPLICATIONS

Bilirubin gallstones, hemolytic crisis, aplastic crisis, and megaloblastic crisis.



TREATMENT



Daily folic acid supplements (1 mg/day).

Severe cases: **Leukocyte-depleted packed red cell transfusions** or **splenectomy** for a clinical cure.

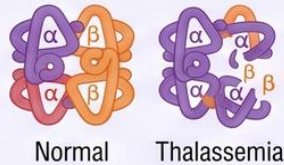
Prophylactic antibiotics are mandatory until age 5 or 2 years post-splenectomy to avoid overwhelming infections.



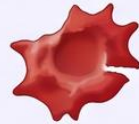
UNDERSTANDING THALASSEMIA SYNDROMES: PATHOGENESIS, TYPES, & MANAGEMENT

DEFINITION & PATHOGENESIS

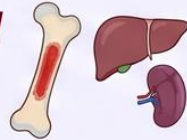
THALASSEMIA SYNDROMES are **CHRONIC HEMOLYTIC ANEMIAS** due to an **INHERITED DEFECT** in globin chain synthesis (*alpha* or *beta* chains), leading to **CHAIN IMBALANCE**.



INEFFECTIVE ERYTHROPOIESIS damages **RBC membranes**.



Anemia drives massive **BONE MARROW EXPANSION** with **EXTRAMEDULLARY HEMATOPOIESIS** in **Liver & Spleen**.



BETA-THALASSEMIA INTERMEDIA & MINOR

INTERMEDIA

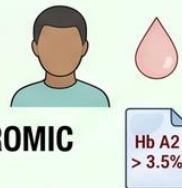
MILDER non-transfusion-dependent anemia, **NO SYMPTOMS** before age 2.



MINOR (TRAIT)

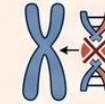
HETEROZYGOUS, COMPLETELY ASYMPTOMATIC

Presents with **MILD MICROCYTIC HYPOCHROMIC ANEMIA** and **ELEVATED Hb A2 (>3.5%)**.



BETA-THALASSEMIA MAJOR (COOLEY ANEMIA)

POINT MUTATIONS in both **beta-globin genes** (chr. 11) result in **ABSENCE** of **beta-globin** production.



Clinical Picture

PROGRESSIVE ANEMIA, WEAKNESS, POOR GROWTH (2nd 6 months of life). Marked **HEPATOSPLENOMEGALY**.



"THALASSEMIC FACIES"

(frontal bossing, flat nasal bridge, maxilla hyperplasia)



SEVERE IRON OVERLOAD leads to **HEMOSIDEROSIS** (heart, liver, endocrine glands).



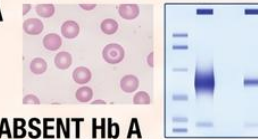
Laboratory Investigations

MICROCYTIC HYPOCHROMIC ANEMIA (low MCV, reticulocytosis).

Peripheral smear: **TARGET CELLS, ANISOCYTOSIS, POIKILOCYTOSIS**

Hb Electrophoresis: **INCREASED Hb F, ABSENT Hb A**

Biochemistry: **INCREASED indirect bilirubin**, high serum ferritin, high serum iron, high transferrin saturation.



Treatment

CHRONIC BLOOD TRANSFUSIONS

(every 3-5 weeks, Hb 9-10 g/dL).

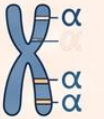
CHELATOR THERAPY (e.g., Deferoxamine, Deferiprone) for serum ferritin > 1000 ng/mL.



CURATIVE THERAPY: HSCT (children <14y, no excessive iron). **SPLENECTOMY** if high transfusion needs or **HYPERSPLENISM**.

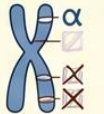


Deficient **ALPHA-CHAIN** synthesis due to **GENE DELETION**.



ALPHA-THALASSEMIA

Deficient **ALPHA-CHAIN** synthesis due to **GENE DELETION**.



1. **SILENT CARRIER** (1 gene deleted): **ASYMPTOMATIC**



2. **ALPHA THALASSEMIA TRAIT** (2 genes deleted): **MILD ANEMIA**



3. **Hb H DISEASE** (3 genes deleted): **HEMOLYTIC ANEMIA** noticed at birth.



4. **HYDROPS FETALIS** (4 genes deleted): **SEVERE HEMOLYSIS** in *utero*, leading to **DEATH**



UNDERSTANDING G6PD DEFICIENCY: A Comprehensive Guide

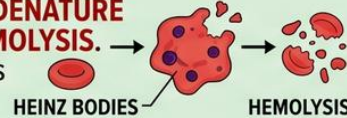
DEFINITION & GENETICS

G6PD (Glucose-6-phosphate dehydrogenase) **DEFICIENCY** is an **INHERITED DISORDER** caused by a **GENETIC DEFECT** in the RBC enzyme **G6PD**. It generates **NADPH**, **PROTECTING** RBCs from **OXIDATIVE INJURY**. It is the **MOST COMMON ENZYMATIC DISORDER** of RBCs, inherited via **SEX-LINKED RECESSIVE MODE** on the **X CHROMOSOME**. Fully expresses in **HEMIZYGOUS MALES** and **HOMOZYGOUS FEMALES**.



PATHOPHYSIOLOGY

In G6PD-deficient individuals, **OXIDATIVE STRESSES DENATURE HEMOGLOBIN**, causing **ACUTE INTRAVASCULAR HEMOLYSIS**. Denatured hemoglobin aggregates are **VISUALIZED** as **HEINZ BODIES** in peripheral blood smears.



TRIGGERING FACTORS OF HEMOLYSIS

Triggers for **ACUTE HEMOLYTIC EPISODES** include:



MEDICATIONS: Sulfonamides, fluoroquinolones (e.g., ciprofloxacin), trimethoprim-sulfamethoxazole, nitrofurantoin, chloramphenicol, antimalarials (primaquine, chloroquine), aspirin, rasburicase.



CHEMICALS: Benzene, naphthalene.



ACUTE ILLNESSES: Bacterial/viral infections (sepsis, hepatitis), DKA.



FOODS: Ingestion of **FAVA BEANS** (FAVISM).

CLINICAL MANIFESTATIONS

Acute intravascular hemolytic episodes **BEGIN 24-72 HOURS POST-EXPOSURE** to oxidant stress.

At **2-4 DAYS**, sudden **ONSET** of **JAUNDICE**, **PALLOR**, and **DARK** or **TEA-COLORED URINE**, with **ABRUPT HEMOGLOBIN FALL** (3-4 g/dL).

SYMPTOMS: abdominal/back pain, weakness, tachycardia.

Can manifest as **SEVERE NEONATAL JAUNDICE** requiring management.



LABORATORY INVESTIGATIONS

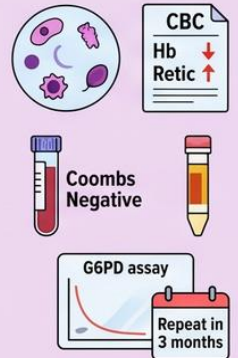
CBC: **LOW HEMOGLOBIN**, **MARKEDLY ELEVATED RETICULOCYTE COUNT**.

Peripheral Blood Smear: shows **BITE CELLS**, **BLISTER CELLS**, **HEINZ BODIES**.

Direct Antiglobulin Test (Coombs): **NEGATIVE**.

Urinalysis: confirms **FREE HEMOGLOBINURIA**.

G6PD Enzyme Activity Assay: confirms **REDUCED LEVELS**, but **CAN BE FALSE-NEGATIVE** (appear normal) during acute episodes due to high reticulocytes; **MUST REPEAT** testing **3 MONTHS LATER**.



TREATMENT & MANAGEMENT

PREVENTION: **STRICT AVOIDANCE** of triggers (oxidative foods, drugs, chemicals).

SUPPORTIVE CARE: **BLOOD TRANSFUSIONS** for **SEVERE ANEMIA** and **OXYGEN THERAPY**.

NEONATAL JAUNDICE is managed with **PHOTOTHERAPY** or **EXCHANGE TRANSFUSION**.



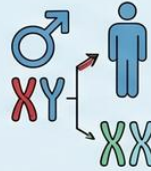
UNDERSTANDING HEMOPHILIA

DEFINITION & CLASSIFICATIONS



Hemophilia: Inherited bleeding disorder due to **coagulation factor** deficiency.

- **Hemophilia A (Classic):** Factor **VIII** deficiency. (X-linked recessive, affects males)
- **Hemophilia B (Christmas Disease):** Factor **IX** deficiency. (X-linked recessive, affects males)
- **Hemophilia C:** Factor **XI** deficiency. (Autosomal recessive, **rarest**)



SEVERITY LEVELS

- **Severe:** Factor activity **<1%** of normal. Frequent **spontaneous bleeding** into joints/muscles.
- **Moderate:** Factor activity **1-5%** of normal. Occasional spontaneous bleeding, **prolonged bleeding** with minor trauma/surgery.
- **Mild:** Factor activity **5% to <40%** of normal. Bleeding **only with major trauma** or surgery.



CLINICAL PRESENTATION



- **Newborns & Infants:** Bleeding from umbilical stump, **intracranial hemorrhage**, cephalohematoma, prolonged post-circumcision/IM injections. **Forehead hematomas** common.
- **Mobile Children & Adolescents:** Widespread bruising, musculoskeletal bleeding.
 - **Hemarthrosis (Joint Bleeding):** **Most common** site (up to 80% of hemorrhages). Affects **ankles, knees, elbows**.
 - **Symptoms:** Tingling/bubbling, stiffness, warmth, **severe pain**, swelling with tense skin.



COMPLICATIONS

- **Chronic joint destruction** and deformity.
- **Muscle atrophy**
- **Transfusion-transmitted viral infections** (HIV, hepatitis B & C).
- **Inhibitors (Major Complication):** **IgG antibodies** that **neutralize clotting factor concentrates**. Occurs in 30% of severe Hemophilia A.



INVESTIGATIONS



- **Coagulation time & Activated partial thromboplastin time (aPTT):** Prolonged
- **Bleeding time & Prothrombin time (PT):** Normal
- **Platelet count:** Normal
- **Factor VIII or IX assays:** Confirm activity levels **<40%**



MANAGEMENT

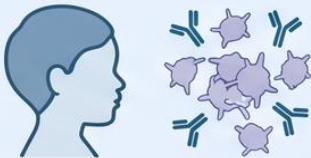
- **Avoidance:** Trauma, IM injections, deep venipunctures, antiplatelet drugs (aspirin, NSAIDs).
- **Joint & Muscle Bleeding (PRICE Protocol):** **P**rotection, **R**est, **I**ce, **C**ompression, **E**levation
- **Specific Replacement Therapy:**
 - **Hemophilia A:** Regular IV **Factor VIII concentrates** (half-life 8-12 hrs).
 - **Hemophilia B:** Regular IV **Factor IX concentrates** (half-life 12-24 hrs).
 - **Alternatives:** FFP, Cryoprecipitate
- **Adjuvant Therapies:** **Desmopressin (DDAVP)** for mild Hemophilia A; Antifibrinolytic drugs like **Tranexamic acid**



IMMUNE THROMBOCYTOPENIA (ITP) IN CHILDHOOD

Acquired Autoimmune Disorder: Low Platelet Count, Destruction, & Impaired Production.

DEFINITION & EPIDEMIOLOGY



Acquired **AUTOIMMUNE** disorder characterized by **LOW PLATELET COUNT** (Isolated Thrombocytopenia $<100,000/\mu\text{mL}$) with **NORMAL WBCs & HEMOGLOBIN**.

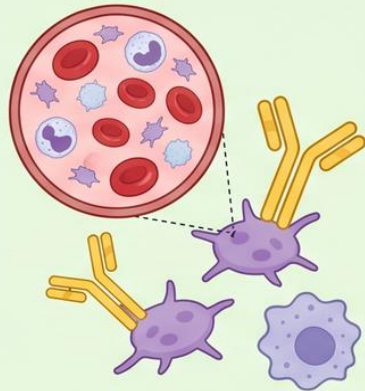
Peak Occurrence:
2-5 YEARS of age.



80% of cases **RESOLVE SPONTANEOUSLY** within 12 months.



PATHOPHYSIOLOGY



AUTOANTIBODIES (typically **IgG**) target platelet membrane antigens (e.g., **glycoprotein IIb/IIIa complex**).

Antibody-coated platelets are **CLEARED RAPIDLY** by **TISSUE MACROPHAGES**, mainly in the **SPLEEN**.

CLINICAL PICTURE



SUDDEN APPEARANCE of **PETECHIAL RASH, PURPURA, & ECCHYMOSES (BRUISING)** in an otherwise healthy child.



Frequently **FOLLOWS VIRAL INFECTIONS** (Rubella, Chickenpox, EBV) by **1-4 WEEKS**.

Physical Exam: **NO FEVER, NO LYMPHADENOPATHY, NO HEPATOSPLENOMEGALY**

More than **HALF: CUTANEOUS BLEEDING** ("dry" purpura).
40%: MUCOSAL BLEEDING ("wet" bleeding) affecting **NOSE, GUMS, BUCCAL AREAS**.

DIAGNOSIS

DIAGNOSIS OF EXCLUSION



CBC shows ISOLATED THROMBOCYTOPENIA



Peripheral Blood Film: **NORMAL TO LARGE PLATELETS** with **NORMAL RBC & WBC MORPHOLOGY**



BONE MARROW ASPIRATION is **UNNECESSARY** in typical cases. If performed: **NORMAL ERYTHROCYTIC/GRANULOCYTIC SERIES** with **NORMAL or INCREASED MEGAKARYOCYTES**

TREATMENT



Goal of therapy is to achieve a platelet count associated with **ADEQUATE HEMOSTASIS, NOT A NORMAL COUNT**.



OBSERVATION ONLY recommended for patients with platelet counts $>20,000/\text{mm}^3$ and **MINOR PURPURA**.

FIRST-LINE MEDICAL THERAPIES for Platelet Counts $<20,000/\text{mm}^3$ plus **SIGNIFICANT MUCOSAL BLEEDING**:



GLUCOCORTICOIDs: Oral Prednisone (2-4 mg/kg/day for 5-7 days) or IV Methylprednisolone (30 mg/kg/day for 2-3 days) to inhibit phagocytosis and antibody synthesis.



INTRAVENOUS IMMUNOGLOBULIN (IVIG): Slows platelet destruction rapidly; 0.8 to 1 g/kg for 1-2 days.



ANTI-D IMMUNE GLOBULIN: For Rh-positive patients at 50-75 $\mu\text{g/kg IV}$.



LIFE-THREATENING BLEEDING (like ICH) is the **ONLY** indication for **IMMEDIATE PLATELET TRANSFUSIONS** alongside high-dose multi-modality therapies.

SECOND-LINE THERAPIES:

TPO receptor agonists (eltrombopag), Rituximab, and Splenectomy.

CHILDHOOD ACUTE LYMPHOBLASTIC LEUKEMIA (ALL)



1. DEFINITION & EPIDEMIOLOGY

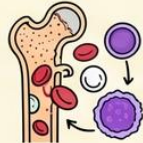
Leukemias are the **most common malignant neoplasms** in childhood, accounting for **31%** of all pediatric malignancies. **Acute Lymphoblastic Leukemia (ALL)** represents **77%** of these childhood leukemia cases, characterized by **genetic abnormalities** in a hematopoietic cell that give rise to an **unregulated clonal proliferation** of **lymphoblasts**. It has a **peak incidence** at 2-3 years of age and is **more common** in **males**.

2. CLINICAL MANIFESTATIONS



Initial presentation:

Nonspecific signs including **anorexia**, **fatigue**, **malaise**, **irritability**, and **intermittent low-grade fever**.



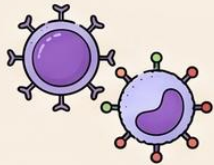
Bone marrow infiltration:

Causes **progressive bone or joint pain** in the lower extremities. Marrow failure leads to **anemia** (pallor, fatigue, exercise intolerance), **thrombocytopenia** (bruising, purpura, petechiae, oral mucosal bleeding, epistaxis), and **leukopenia** (fever and susceptibility to infections).



Organ infiltration:

Causes **generalized lymphadenopathy**, **hepatosplenomegaly**, **testicular enlargement**, or **central nervous system (CNS) involvement** manifesting as **cranial neuropathies**, **headache**, and **seizures**. **Airway compression** by mediastinal nodes can cause respiratory distress.



3. CELLULAR CLASSIFICATION

Based on cell membrane markers, **85%** of cases are **B-lymphoblastic leukemia** and **15%** are **T-lymphoblastic leukemia**.

5. PROGNOSTIC FACTORS FOR ALL



Favorable Factors:

Age between 1 and 9 years;
Initial white blood cell (WBC) count less than $50 \times 10^9/L$;
B-precursor immunophenotype;
Female sex.



Unfavorable Factors:

Age less than 1 or more than 10 years;
WBC count above $50 \times 10^9/L$;
T-cell immunophenotype; Male sex;
Presence of CNS infiltration at diagnosis.

4. DIAGNOSIS



Peripheral blood findings show **anemia**, **thrombocytopenia**, and **circulating lymphoblasts**.

Bone marrow aspiration and biopsy are confirmative and diagnostic if more than 25% of the bone marrow cells are lymphoblasts.

CSF examination defines **CNS leukemia** if there are more than 5 WBCs/ μL with visible blasts.

Blood chemistry shows markedly elevated **lactate dehydrogenase (LDH)** and **uric acid**.

6. TREATMENT & SUPPORTIVE CARE



Multi-phase chemotherapeutic regimens include **Remission induction**, **Consolidation**, and **Maintenance** phases.

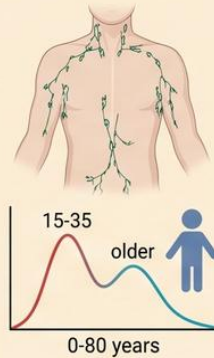
Supportive care is vital to manage **Tumor Lysis Syndrome (TLS)**, a life-threatening complication managed with **aggressive IV hydration** before chemotherapy, **uric acid reduction** via **allopurinol**, and **close electrolyte monitoring**.

RBC and platelet transfusions, alongside **prophylactic antimicrobials**, are standard supportive measures.

HODGKIN LYMPHOMA (HL): AN EXPLANATORY OVERVIEW

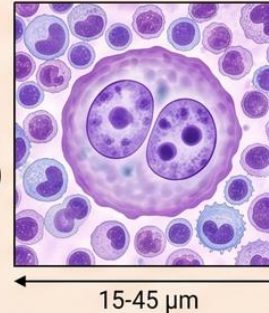
DEFINITION & EPIDEMIOLOGY

Hodgkin lymphoma (HL) is a malignant process involving the lymphoreticular system that accounts for **6% of childhood cancers**. It shows a **biconcave bimodal age distribution** peaking at **15-35 years**, with a notable **male predominance** among young children.



PATHOGNOMONIC FEATURE

The **Reed-Sternberg (RS) cell** is the hallmark diagnostic feature, characterized as a large cell (15-45 μm in diameter) containing multiple or multilobulated nuclei ("**owl-eye**" appearance).



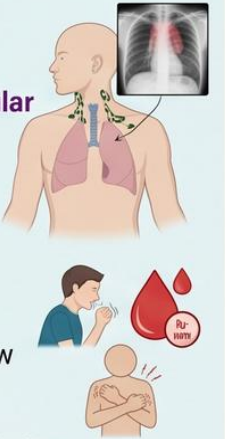
CLINICAL MANIFESTATIONS

Presents primarily with **painless, non-tender, firm, rubbery lymphadenopathy**, most commonly affecting the **cervical or supraclavicular** lymph node chains.

Mediastinal node involvement can cause airway obstruction presenting as **dyspnea, hypoxia, and a chronic cough**.

Advanced disease causes hepatosplenomegaly, pleural or pericardial effusions, and bone marrow infiltration leading to **anemia, neutropenia, or thrombocytopenia**.

Pruritus is a less common systemic finding.



STAGING & B SYMPTOMS

HL is categorized based on the presence of **systemic constitutional symptoms**.

"A" Symptoms: Signifies the complete **lack of systemic constitutional symptoms**.

"B" Symptoms: Fulfilling at least one of the following criteria, which carries **significant prognostic importance**.

- **Unexplained persistent fever higher than 38°C**
- **Involuntary weight loss of more than 10% of total body weight** over the preceding 6 months; or
- **Drenching night sweats**



DIAGNOSIS

Excisional lymph node biopsy is strictly preferred over needle biopsy to ensure adequate tissue structure for light microscopy and immunohistochemical studies.

Initial staging workup includes chest radiographs, **CT scans** (neck, chest, abdomen, cr chest, abdomen, pelvis), alongside a **PET scan**.

Laboratory studies include **CBC, blood film, ESR, serum ferritin** (prognostic marker), and **bone marrow aspiration/biopsy** to rule out advanced stage 4 disease.



TREATMENT

Risk-adapted therapy utilizing multi-agent **combined chemotherapy** with or without **radiation therapy** and **immunotherapy**.

Standard chemotherapeutic regimens are based on **ABVD** (doxorubicin, bleomycin, vinblastine, dacarbazine) or **COPP** (cyclophosphamide, vincristine, procarbazine, prednisone).

Targeted agents include **anti-CD20 Rituximab** and **anti-CD30 Brentuximab vedotin** to target RS cells.



CHAPTER 07

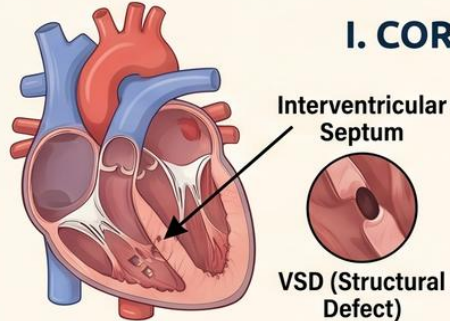
VII. CVS



ACYANOTIC HEART: VSD (PART 1) • ACYANOTIC HEART: VSD (PART 2) • ACYANOTIC HEART: ASD • ACYANOTIC HEART: PDA • CYANOTIC HEART: TGA • CYANOTIC HEART: FALLOT / TOF
• HEART FAILURE

VENTRICULAR SEPTAL DEFECT (VSD):

PART 1 - Anatomy & Hemodynamics



I. CORE DEFINITION & PREVALENCE

Keypoint 1:

Concept: A **structural defect** in the **interventricular septum**.

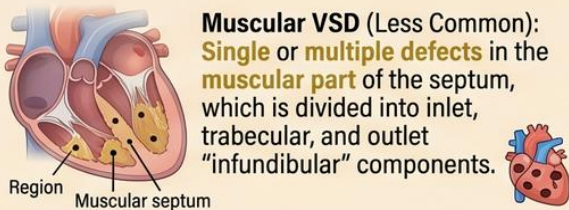
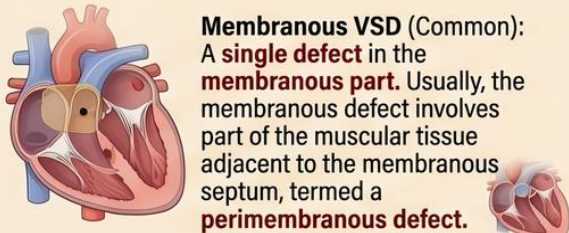
Keypoint 2:

Prevalence: VSD is the **most common form** of congenital heart disease (CHD) and accounts for about **30% of all such defects**.

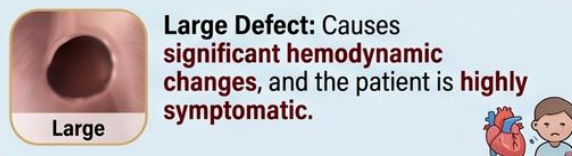
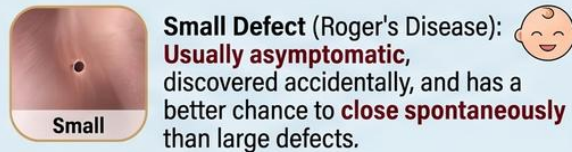


II. ANATOMICAL AND SIZE CLASSIFICATION

A. According to the Site of the Defect



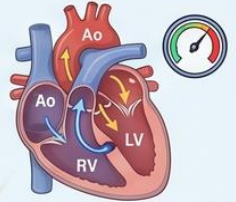
B. According to the Size of the Defect



III. COMPLEX HEMODYNAMICS

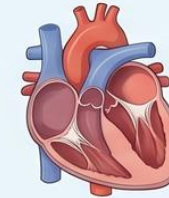
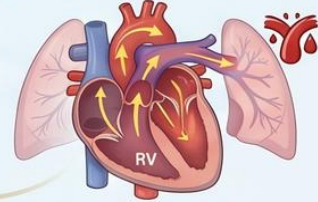
Keypoint 1:

Shunt Mechanism: Blood from the **left ventricle (LV)** is **divided into 2 portions**: one portion passes to the aorta and the other passes **through the defect to the right ventricle (RV)**. The amount of shunted blood depends on the **size of the VSD** and the level of **pulmonary vascular resistance**.



Keypoint 2:

Pulmonary Plethora: **All this blood** (RV output & part from LV output that came through VSD) passes through the **pulmonary artery**, leading to **pulmonary plethora**.



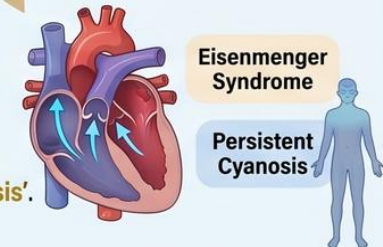
Keypoint 3:

Chamber Dilatation: This blood passes to the **left atrium (LA)**, leading to **left atrium dilatation** and subsequent **left ventricular hypertrophy (LVH)** or **left ventricular dilatation**.



Eisenmenger Syndrome:

Persistence of pulmonary plethora causes **pulmonary hypertension**, leading to symptoms of **low cardiac output (COP)**. Pulmonary hypertension **increases right ventricular pressure**, causing Eisenmenger Syndrome and resulting in '**persistent cyanosis**'.



VENTRICULAR SEPTAL DEFECT (VSD): CLINICAL PICTURE & MANAGEMENT (PART 2)

IV. CLINICAL MANIFESTATIONS

A. Symptoms



Small VSD: Asymptomatic, normal growth & development.



Moderate to Large VSD: Delayed growth, tachypnea & excessive sweating (feeding), repeated pulmonary infections, CHF symptoms (infancy). Exertional dyspnea (older children).



Eisenmenger's Phase: Cyanosis, clubbing, decreased activity (with pulmonary vascular obstructive disease).

B. Signs



Inspection & Palpation: Precordial bulge, hyperkinetic apex, systolic thrill (lower left sternal border) with large VSD.



Auscultation (Heart Sounds): P2 intensity normal (small shunt) / increased (large shunt). S2 loud & single (pulmonary hypertension).



Auscultation (Murmurs): Harsh pansystolic murmur (lower left sternal border, propagated). Sometimes early systolic. **Infundibular VSD:** soft early diastolic AR murmur (aortic cusp herniation).

V. INVESTIGATIONS



Electrocardiography (ECG): Normal (small). LVH & LAH (moderate). BVH +/- LAH (large). RVH only (pulmonary vascular obstructive disease).



Radiography (CXR): Cardiomegaly (LA, LV, sometimes RV). Increased pulmonary vascular markings. Relates to shunt magnitude.

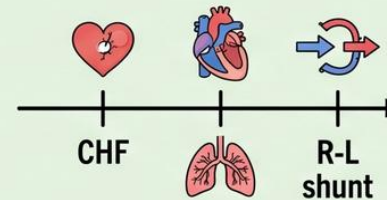


Echocardiography: Identifies defect (number, size, location), estimates PA pressure, detects chamber enlargement, associated defects, shunt magnitude.

VI. NATURAL HISTORY



Spontaneous Closure: More frequent with small defects & first 6 months. ~60% muscular (by 8 yrs), ~35% perimembranous (by 5 yrs). Inlet/outlet do not close.



Complications Timeline: CHF (large VSDs, 6-8 weeks). Pulmonary vascular obstructive disease (large VSDs, as early as 6-12 months, R-L shunt usually teenage years).

VII. MANAGEMENT



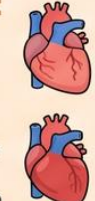
Medical Management: Diuretics & ACE inhibitors (CHF, 2-4 months for growth). High-calorie feedings (NG tube/oral). No exercise restriction (no hypertension).



Definitive Closure Indications & Timing: Failure to thrive. PA pressure >50% systemic (by 1 year). Dilated left chambers, intractable CHF, subaortic VSD with AR.



Procedures: Nonsurgical device closure (selected muscular). PA banding (palliative, rarely used). Surgical closure (patch closure under bypass).



ATRIAL SEPTAL DEFECT (ASD)

I. DEFINITION & PREVALENCE

Concept: Structural defect in the inter-atrial septum.

Prevalence

Isolated ostium secundum defect occurs in **5% to 10%** of all congenital heart defects (CHDs).

Gender Ratio: Significantly **more common** in **females** than in males, with a male-to-female ratio of **1:2**.

II. ANATOMICAL TYPES

Ostium Secundum Defect

Commonest presentation, located in the **center** of the septum.

Ostium Primum Defect

Located in the **lower portion** of the septum.

Sinus Venosus Defect

Located high up near **superior vena cava entry**, or lower down near **inferior vena cava**.

Coronary Sinus ASD

Located within the **roof of the coronary sinus**.

III. HEMODYNAMICS

Shunt Mechanism

Left atrial pressure is **physiologically higher** than right atrial pressure. Therefore, blood is **continuously shunted** from the left atrium (LA) to the right atrium (RA).

Volume Overload

The RA receives extra shunted volume and normal venous return, leading directly to **right atrial dilatation and enlargement**.

Pulmonary Consequences

Large volume propelled into **right ventricle (RV)**, causing marked **RV volume overload and dilatation**. Blood exits to **pulmonary artery**, producing **pulmonary artery enlargement**, lung plethora, and repeated chest infections.

IV. CLINICAL MANIFESTATIONS

A. Symptoms

Small ASD

Infants **well-developed** and remain **completely asymptomatic**.

Large ASD

Infants may present with **poor weight gain** or show **progressive symptoms and signs of CHF**.

B. Physical Signs

Inspection & Palpation

Large left-to-right shunt: **visible precordial bulge**, **prominent RV cardiac impulse**, **palpable PA pulsations**.

Heart Sounds

The second heart sound (S_2) is **widely split and fixed**, which stands as the **pathognomonic and characteristic finding** of ASD.

Murmurs

Soft (grade 2 to 3/6) **systolic ejection murmur** best audible over pulmonary area (functional murmur due to relative pulmonary stenosis). With large shunt: **mid-diastolic rumble** at lower left sternal border (relative tricuspid stenosis).

Crucial Note

There is absolutely **NO MURMUR** produced by the actual passage of blood through the ASD itself.

V. INVESTIGATIONS

Electrocardiography (ECG)

Typical findings show **right axis deviation (RAD)** accompanied by **mild RVH** or a **right bundle branch block (RBBB)** pattern in lead V_1 .

Chest Radiography (CXR)

Reveals **distinct cardiomegaly** with clear **enlargement of RA and RV**. **Prominent pulmonary artery (PA) segment** and **increased pulmonary vascular markings** are visible with significant shunt.

V. INVESTIGATIONS

Electrocardiography (ECG)

Typical findings show **right axis deviation (RAD)** accompanied by **mild RVH** or a **right bundle branch block (RBBB)** pattern in lead V_1 .

Chest Radiography (CXR)

Reveals **distinct cardiomegaly** with clear **enlargement of RA and RV**. **Prominent pulmonary artery (PA) segment** and **increased pulmonary vascular markings** are visible with significant shunt.

Echocardiography

Definitive diagnostic modality; identifies exact **anatomical type and size**, estimates **PA pressure**, detects **chamber enlargement**, rules out associated defects, and estimates **shunt magnitude (Q_p/Q_s)**.

VI. MANAGEMENT

Exercise Guidelines

Absolutely **no exercise restriction** is necessary.

Medical Management

Recommended in infants with **CHF** due to high success rate and possibility of **spontaneous defect closure**.

Definitive Closure Timing

Considering spontaneous closure, ASDs are preferably closed when children reach **2-4 years**.

Early Closure Indications

Closure indicated **before 2 years** if:

- Significant left-to-right shunt with **clear evidence of RV volume overload** (dilated right chambers or $Q_p/Q_s \geq 1.5:1$);
- Failure to thrive (FTT)**;
- Congestive heart failure** completely unresponsive to medical management.

Nonsurgical Device Closure

Transcatheter closure using catheter-delivered device is the **absolute standard preferred method**, applicable only for **secundum ASD**.

Surgical Closure

Strictly reserved for patients where device closure is **not anatomically appropriate**. Performed via **simple suture or patch closure** under cardboard bypass.

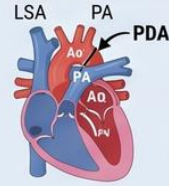
PATENT DUCTUS ARTERIOSUS (PDA)

EXPLANATORY GUIDE

I. DEFINITION & PREVALENCE

Concept

It is the **persistence** of the normal fetal communication between the **main pulmonary artery** and the **aortic arch**, located specifically **after the origin of the left subclavian artery**.



Prevalence

Occurs in **5% to 10%** of all CHDs when excluding premature infants.



Gender Ratio

It is significantly more common in females than in males, carrying a **male-to-female ratio of 1:3**. It is a highly common and major clinical problem in **premature infants**.



II. HEMODYNAMICS

Shunt Mechanism

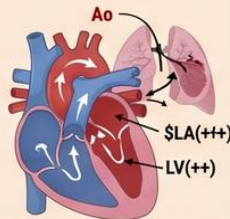
Pressure in the systemic aorta remains **higher** than the pressure in the pulmonary artery during both systole and diastole.



Chamber Overload

Consequently, blood is continuously shunted from the aorta into the pulmonary artery, which simultaneously receives normal venous output from the RV.

This leads directly to **increased pulmonary blood flow**, lung plethora, **left atrial dilatation (LA++)**, and **left ventricular dilatation/hypertrophy (LV++)**, creating a physiological circuit very similar to VSD.



III. CLINICAL MANIFESTATIONS

A. Symptoms



Small PDA

The patient is completely **asymptomatic**.



Large PDA

Frequently causes **poor weight gain**, decreased exercise tolerance, **repeated lower respiratory tract infections**, and **congestive heart failure (CHF)** during. Older children may present with **exertional dyspnea**.



B. Physical Signs

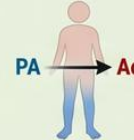
General Examination

Infants with secondary CHF show prominent **tachycardia** and **tachypnea**. Large PDAs present with **bounding peripheral pulses** and a classic **water hammer pulse**, caused directly by a wide pulse pressure with elevated systolic pressure and lower diastolic pressure.



Differential Cyanosis Risk

If pulmonary vascular obstructive disease develops, it induces a **right-to-left ductal shunt**, resulting in **cyanosis strictly in the lower half of the body** (differential cyanosis).



Inspection & Palpation

Large shunts present with a **hyperactive precordium**. A **palpable systolic thrill** may be present at the **upper left sternal border** in the left infraclavicular area.



Auscultation (Heart Sounds)

P_2 is usually normal but its intensity becomes **markedly accentuated** if pulmonary hypertension is present.



Auscultation (Murmur)

Characterized by a pathognomonic grade 1 to 4 of 6 **continuous machinery murmur**, best audible at the **left infraclavicular area** and propagated to the pulmonary area and the back.



IV. INVESTIGATIONS

Electrocardiography (ECG)

Findings are highly similar to VSD. A normal ECG or **isolated LVH** is seen with a small-to-moderate PDA, whereas **biventricular hypertrophy (BVH)** is characteristically seen with large PDAs.



Chest Radiography (CXR)

Radiographs are similar to VSD. They may be completely normal with a small-shunt PDA. Moderate-to-large shunts reveal varying degrees of **cardiomegaly** with specific enlargement of the **LA, LV, and ascending aorta**, accompanied by **increased pulmonary vascular markings**. If pulmonary vascular obstructive disease develops, heart size normalizes but displays a marked prominence of the **PA segment and hilar vessels**.



Echocardiography

Assesses the **exact size, site, and shape** of the PDA, and documents the **direction of flow** through the shunt. The measured dimensions of the LA and LV provide an indirect, objective assessment of the **magnitude of the left-to-right ductal shunt**.



V. NATURAL HISTORY & MANAGEMENT

B. Management Protocol

Medical Prophylaxis (Premature Only)

Pharmacological closure using **indomethacin, ibuprofen, or paracetamol** is highly effective and widely used to close the PDA **strictly in premature infants**.

Exercise Rules

No exercise restriction is needed in the absence of pulmonary hypertension.

Definitive Closure Indication

Establishing a **diagnosis of PDA** in a term infant is a **direct indication for its closure**, except in a completely silent PDA with no hemodynamic consequences.

Nonsurgical Intervention

Nonsurgical device occlusion of the PDA via **cardiac catheterization** has become the **standard of care**.

Surgical Intervention

Surgical closure (ligation or patch) is reserved **exclusively** for patients in whom device closure is not applicable.



COMPLETE TRANSPOSITION OF THE GREAT ARTERIES (TGA)

I. PREVALENCE & CORE DEFINITION

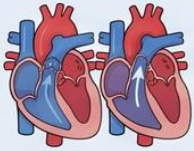


PREVALENCE
Most common **cyanotic** disorder in neonates

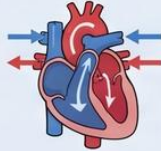
GENDER DISPARITY
More frequent in **males**
MALE-TO-FEMALE RATIO: 3:1



ANATOMICAL REVERSAL
Reversal of normal anteroposterior relation



NORMAL
Aorta **POSTERIOR & RIGHTWARD**
→ **ANTERIOR & LEFTWARD**



TGA
Aorta **ANTERIOR & RIGHTWARD**
→ **POSTERIOR & LEFTWARD**

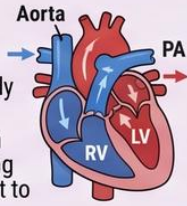
II. PATHOPHYSIOLOGY & PARALLEL CIRCULATIONS

INDEPENDENT CIRCUITS
Two independent circulations in **PARALLEL**, not series



ABNORMAL ORIGINS

- **Aorta** arises from **RIGHT VENTRICLE (RV)**, carrying **desaturated (blue)** blood directly to the body
- **Pulmonary Artery** arises from **LEFT VENTRICLE (LV)**, carrying **oxygenated (red)** blood straight to the lungs



SURVIVAL OBLIGATION
INCOMPATIBLE WITH LIFE without mixing

MIXING SITES
Essential for survival via **PFO, ASD, VSD, or PDA**



ANATOMICAL SUBTYPES
60% Intact Ventricular Septum (Atrial shunt only)



ANATOMICAL SUBTYPES

- **60% Intact Ventricular Septum** (Atrial shunt only)
- **40% VSD Present**

III. CLINICAL MANIFESTATIONS

A. HISTORY & SYMPTOMS



CYANOSIS TIMELINE
Intense **cyanosis since birth**



NEONATAL DISTRESS
Rapid CHF symptoms, progressive dyspnea, severe feeding difficulties



SEPTAL INFLUENCE
Intact Septum = Symptoms within **2 DAYS**, more intense cyanosis than VSD cases



B. PHYSICAL EXAMINATION SIGNS



CYANOSIS QUALITY
NOT RELIEVED by 100% O₂ therapy



AUSCULTATION (MURMURS)

- **Intact Septum:** No murmur or soft, non-specific
- **Associated VSD:** Classic **HARSH MURMUR**
- **Associated PS:** **DISTINCTIVE EJECTION SYSTOLIC MURMUR**



SYSTEMIC CONGESTION
Noticeable **HEPATOMEGALY**, worsening dyspnea (if CHF)

IV. DIAGNOSTIC INVESTIGATIONS



ELECTROCARDIOGRAM (ECG):

- Shows **RIGHT-AXIS DEVIATION (RAD)** and **RIGHT VENTRICULAR HYPERTROPHY (RVH)** due to systemic RV pressure

CHEST X-RAY (CXR):

- Generalized **CARDIOMEGALY**
- Increased **PULMONARY VASCULAR MARKINGS**
- Classic **EGG-SHAPED** ("EGG ON A STRING")



ECHOCARDIOGRAM
DEFINITIVE, PRIMARY DIAGNOSTIC MODALITY

V. MANAGEMENT PROTOCOL

A. PALLIATIVE EMERGENCY PROCEDURES

- Improve mixing to prevent death from severe hypoxia & metabolic acidosis
- **INTRAVENOUS PROSTAGLANDIN (\$PGE_1)**:
- **IMMEDIATE** administration to keep **DUCTUS ARTERIOSUS PATENT**
- **RASHKIND BALLOON ATRIAL SEPTOSTOMY**
- Creates or enlarges **ATRIAL SEPTAL DEFECT** for mixing



B. DEFINITIVE CORRECTIVE OPERATIONS

- **ARTERIAL SWITCH OPERATION (ASO):**
- **PROCEDURE OF CHOICE**
Surgically switches Aorta and Pulmonary Artery to correct ventricles. Must occur **EARLY (3-8 WEEKS)**
- **ATRIAL (VENOUS) SWITCH**
Historical (Mustard/Senning) redirects venous inflows
- **VENTRICULAR SWITCH (RASTELLI OPERATION)**
For **COMPLEX PATIENTS (VSD + PS)**



TETRALOGY OF FALLOT (TOF) & HYPERCYANOTIC SPELLS (SUMMARY)

I. TOF CORE PATHOPHYSIOLOGY

Definition: Most common cyanotic disorder in children.

The 4 Classic Components:

1. Ventricular Septal Defect (VSD).
2. Overriding Aorta.
3. Pulmonary Stenosis (mainly infundibular).
4. Right Ventricular Hypertrophy (RVH).



Shunt Dynamics:

Right-to-left shunt via VSD causes systemic cyanosis. It varies dynamically with **exercise, emotion, or infundibular muscle spasms.**

II. CLINICAL MANIFESTATIONS

Cyanosis: Presents early, worsening after PDA closure, and **fails to respond to oxygen therapy.**



The Squatting Mechanism:

Older children squat to **increase systemic vascular resistance**, which reduces the right-to-left shunt and improves pulmonary blood flow.

Heart Failure Absence:

Congestive heart failure typically does not occur in TOF.

Auscultation Murmur:

A **systolic ejection murmur** at the left sternal border caused strictly by the **pulmonary stenosis**, not the VSD. The murmur **softens or disappears** during a cyanotic spell.



III. HYPERCYANOTIC (TET) SPELLS

Clinical Picture: Sudden attacks of deepening cyanosis, **tachypnea, irritability, and crying.** Severe spells can lead to **unconsciousness, convulsions, or death.**



Precipitating Factors:

Crying, defecation, fever, feeding, or awakening from naps. Iron deficiency can also trigger it.



Auscultation Murmur:

A **systolic ejection murmur** at the left sternal border caused strictly by the **pulmonary stenosis**, not the VSD. The murmur **softens or disappears** during a cyanotic spell.



IV. DIAGNOSTIC INVESTIGATIONS

ECG:

Shows **right-axis deviation (RAD)** and **right ventricular hypertrophy (RVH).**



Chest X-ray (CXR):

Characteristically reveals a normal heart size with a **boot-shaped contour ("coeur en sabot")** and decreased pulmonary vascular markings.



Echocardiography:

Stands as the **definitive, primary diagnostic modality.**



V. EMERGENCY MANAGEMENT PROTOCOL (TET SPELLS)

Execute these clinical steps immediately and strictly in order:



1. **Knee-Chest Position.**



2. **Calm the Child** (avoid iatrogenic agitation).



3. **Supplemental Oxygen** (via least aggravating method).



4. **Morphine Sulfate** (subcutaneous or slow IV).



5. **Sodium Bicarbonate IV** (to treat acidosis).



6. **Warmed Fluid Bolus** (\$10 ml/kg of 0.9% NaCl IV).



7. **Beta-Blockers** (e.g., Esmolol to reduce infundibular spasm).



8. **Ketamine IV or Phenylephrine IV** (to increase systemic vascular resistance).



9. **Intubation / General Anesthesia.**



10. **Urgent Surgical Shunt.**



CRITICAL ABSOLUTE CONTRAINDICATIONS:

Give **NO INOTROPES** and **NO DIURETICS** during a spell.

VI. SURGICAL TREATMENT

Palliative: Modified Blalock-Taussig shunt.



Definitive: Complete surgical repair, preferred at 6 to 9 months of age.



PEDIATRIC HEART FAILURE (HF)



Definition: A clinical state where the heart cannot deliver an adequate cardiac output to meet the body's metabolic needs.

I. PATHOGENESIS & ETIOLOGY



Core Mechanisms & Causes



Increased Afterload (Pressure Overload): Seen in Aortic Stenosis (AS), Coarctation of the Aorta (CoA), systemic hypertension.



Increased Preload (Volume Overload): Leads to High-Output Failure. Seen in Left-to-Right shunts (VSD, PDA, ASD), valve regurgitation, severe anemia.



Decreased Myocardial Contractility: Seen in Cardiomyopathy, Myocarditis.



Tachyarrhythmias: Shorten diastolic filling time.



Precipitating Factors: Infections (endocarditis, pneumonia), physical/emotional stress, arrhythmias, anemia, iatrogenic factors (excessive IV fluids, beta-blockers in acute stages, corticosteroids causing fluid retention).

II. CLINICAL PICTURE



A. In Infants (The Diagnostic Triad)

The Essential Triad

Tachypnea Tachycardia Enlarged tender liver



Other Manifestations: Poor weight gain, restlessness, irritability, weak cry, feeding difficulties.

B. In Older Children



Left-Sided Heart Failure (Pulmonary Congestion & Low COP):

Low Cardiac Output (COP): Headache, syncope, pale cold skin, peripheral cyanosis, easy fatigability.



Pulmonary Congestion: Dyspnea, cough, hemoptysis, recurrent chest infections, bilateral basal crepitations, pleural effusion.



Cardiovascular Signs: Weak pulse, tachycardia, pulsus alternans, low systolic BP, apex beat shifted out & down (LV enlargement).



Right-Sided Heart Failure (Systemic Congestion):

Systemic Congestion: Congested pulsating neck veins, epigastric/right hypochondrium pain, enlarged tender liver, anorexia, nausea, vomiting, ascites. Lower limb edema is a late sign.



Cardiovascular Signs: Weak & rapid pulse, low systolic BP, apex shifted outwards (RV enlargement).

III. FUNCTIONAL CLASSIFICATION



Ross Classification (For Infants & Children):

Class I No limitations or symptoms.



Class II Mild tachypnea/diaphoresis with feedings; dyspnea on exertion in older children; no growth failure.



Class III Marked tachypnea/diaphoresis with feedings or exertion; prolonged feeding times with growth failure.



Class IV Symptomatic at rest with tachypnea, retractions, grunting, or diaphoresis.



IV. INVESTIGATIONS



Chest X-ray (CXR):

Cardiomegaly, signs of pulmonary edema (fluffy markings), increased pulmonary vascularity with Left-to-Right shunts.



ECG: Chamber dilatation, flat/inverted ST segment/T waves, causative arrhythmias.



Echocardiography:

Primary tool used to definitively assess ventricular function and identify structural causes.

V. GENERAL & NUTRITIONAL MANAGEMENT



Positioning & Oxygen:

Keep patient in a semi-upright position, provide humidified oxygen (40%–50%).



Infant Hypercaloric Rules:

Infants need significantly higher caloric intakes (\$150–160 kcal/kg/day) due to tachypnea/increased work of breathing. Increase caloric density via fortification, provide frequent small feedings. Use nasogastric (NG) feeding if needed.



Older Children:

Strict bed rest, salt restriction, avoidance of salty snacks.

VI. PHARMACOLOGICAL THERAPY



A. Diuretics (Preload Reduction)

Principal agents for congestion, do not improve CO directly.

Loop Diuretics (Furosemide): Absolute drug of choice.



Side Effects: Hypokalemia (premeditates digitalis toxicity), hyponatremia, hypomagnesemia, hypovolemia, hypotension, ototoxicity.

Aldosterone Antagonists (Spironolactone): Added to loop diuretics to prevent hypokalemia & overcome resistance.

Discontinue if ACE inhibitors are used to avoid fatal hyperkalemia.

B. Afterload-Reducing Agents (Vasodilators)



Vital for infants with large Left-to-Right shunts (VSD, PDA) and DCM.

Mixed Vasodilators (ACE Inhibitors - Captopril, Enalapril): Act on both arteriolar and venous beds. Highly popular for chronic severe HF.

C. Inotropic Agents (Myocardial Performance)



Rapid-Acting Inotropes (Dopamine, Dobutamine, Epinephrine): Used strictly in critically/ acutely ill.

Phosphodiesterase Inhibitors (Milrinone): Augment contractility and decrease afterload with less chronotropy.

D. Digitalis Protocol (Digoxin)



Indications: HF with decreased systolic function (e.g., DCM) & atrial arrhythmias. Added later in large shunts.

Absolute Contraindications: Digitalis toxicity, paroxysmal ventricular tachycardia, Obstructive lesions (TOF, Hypertrophic Cardiomyopathy). Avoid co-administration with calcium, adrenaline, or vagolytic atropine.



Age-Based Oral Digitalizing Dosages:

Premature: \$20 mcg/kg **Newborn:** \$30 mcg/kg
< 2 years: \$40–50 mcg/kg **> 2 years:** 30–40 mcg/kg

Administration: details in fnovbn on: 30–40 mcg/kg.

Digitalis Toxicity Symptoms: Anorexia, nausea, vomiting, bradycardia, visual disturbances, severe arrhythmias.

Toxicity Management: Stop digoxin immediately, stop potassium-losing diuretics, give potassium supplements, treat arrhythmias, administer Digitalis specific antibodies.

E. Beta-Adrenergic Blockers (e.g., Carvedilol)



Carvedilol is drug of choice.

Used strictly in stable chronic stages of DCM (Class II–III).

Never use in acute phase; always combine with diuretics and ACE inhibitors.

CHAPTER 08

VIII. CNS



CEREBRAL PALSY • FLOPPY INFANT • DUCHENNE MUSCLE DYSTROPHY • INFECTION

CEREBRAL PALSY (CP): COMPREHENSIVE GUIDE

I. DEFINITION & CORE CHARACTERISTICS

- **Concept:** It is an encephalopathy resulting from malfunction of the motor unit of the developing brain due to a brain insult during prenatal, natal, and postnatal periods.
- **Core Deficit:** Central motor deficit that is strictly **non-progressive**, **non-familial**, and **non-hereditary**.
- **Associated Deficits:** May be associated with mental retardation, epilepsy, visual defect, deafness, and speech defect.



II. ETIOLOGY & TIMING OF INSULT

Prenatal Factors (80%): TORCH infection, teratogens, and intrauterine hypoxia.



Perinatal Factors (10%): TORCH infection, teratogens, intrauterine hypoxia.



Perinatal Factors (10%): Birth asphyxia, birth trauma, and IC hemorrhage.



Postnatal Factors (10%): Kernicterus, meningitis, encephalitis, and IC hemorrhage.

III. CLINICAL CLASSIFICATIONS

A. Topographic (Anatomic) Classification



Hemiplegia: Arm & Leg on same side of body (R or L).



Diplegia: Arms and Legs but legs are more involved.



Quadriplegia or Tetraplegia: Arms and Legs equally involved OR Arms more involved than legs.



Monoplegia: One limb.



Triplegia: Three limbs.



Paraplegia: Only legs involved.

B. Physiological Classification (Major Motor Abnormalities)

Spastic CP (70 to 80%): Hypertonia with hyperreflexia, positive Babinski sign (extensor plantar response), and persistent primitive reflexes.



Dyskinetic CP (10 to 20%): Abnormal movements and/or change in tone (increased or decreased).



- **Rigidity:** Hypertonia with hyporeflexia or normal reflexes.
- **Chorea:** Quick irregular dysrhythmic movements involve proximal skeletal muscles.
- **Athetosis:** Slow twisting movements, usually along long axis of a limb, more prominent distally.
- **Choreoathetosis:** Chorea and Athetosis are commonly seen.



Mixed Forms: Spasticity and ataxia may coexist.

- **Dystonia:** Painful sustained contraction of a muscle group's → prolonged, twisting, posturing of the face, limb or trunk, which involved.



Ataxic CP (5 to 10%): Disturbances in balance and equilibrium: walk unsteadily, poor coordination and balance often with hypotonia, hyporeflexia, nystagmus.



Atonic (Hypotonic) CP: Floppy infant with hyperreflexia, usually with marked mental retardation.



C. Functional Classification



Class 1: No limitation of activity.



Class 2: Slight to moderate limitation.



Class 3: Moderate to great limitation.



Class 4: No useful physical activity.

IV. MULTIDISCIPLINARY MANAGEMENT

The Team: A team of physicians: pediatricians, occupational and physical therapists, social workers, speech pathologists.



The Role of Pediatrician:



Treatment of comorbidities (seizures, malnutrition, spasticity).



Supervising nutritional status.



To prescribe muscles relaxants (for muscles spasticity), sedatives (to reduce sleep and decrease irritability).



To explain with the help of the social workers and psychiatrist the nature of the disease to the family to ensure better cooperation.



Refer to orthopedic surgery to correct deformities, provide stability and rehabilitation.



THE FLOPPY INFANT SYNDROME

I. CORE DEFINITION

Concept: Severe persistent hypotonia present at birth or early infancy.



II. ANATOMICAL & ETIOLOGICAL CAUSES



A. Cerebral Causes



Perinatal Asphyxia: Brain insult leading to central hypotonia.



Kernicterus: Bilirubin-induced encephalopathy.



Cerebral Palsy: Specifically the atonic (hypotonic) type.



Chromosomal Anomalies: Including Down syndrome.



B. Spinal Cord Lesions



Anterior Horn Cell Disease: Werdnig-Hoffman spinal muscular atrophy.



C. Peripheral Nerves



Acute Polyneuropathy: Generalized dysfunction of peripheral nerves.



Guillain-Barré Syndrome (GBS): Post-infectious demyelinating polyneuropathy.



Congenital Sensory Neuropathy: Inherited structural or functional nerve abnormalities.



B. Spinal Cord Lesions



Anterior Horn Cell Disease: Werdnig-Hoffman spinal muscular atrophy.



Early Extensive Poliomyelitis: Viral destruction of lower motor neurons.



D. Neuromuscular Junction



Neonatal Myasthenia Gravis: Transient autoimmune neuromuscular blockade.



Infantile Botulism: Neurotoxin-induced inhibition of acetylcholine release.



E. Myopathies



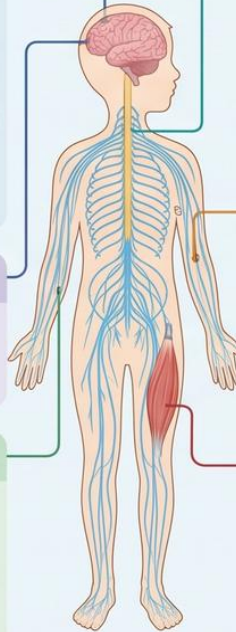
Muscular Dystrophies: Progressive genetic muscle wasting disorders.



Congenital Myopathies: Structural embryonic muscle abnormalities.



Polymyositis: Inflammatory muscle disease.



III. CLINICAL SIGNS & EVALUATION OF HYPOTONIA



Frog Leg Position: Denotes marked hypotonia of the lower limbs.



Head Lag: When the baby is pulled up from his hands while in a supine position, the head lags backwards, denoting hypotonia of the neck muscles.



Curved Trunk (Ventral Suspension): When the baby is suspended in a prone position over the examiner's hand, they drop around it in a "U-shape", denoting hypotonia of the trunk muscles.



Slippage (Vertical Suspension): The infant slips through the examiner's hands when held under the arms, denoting hypotonia of the shoulder girdle muscles.

IV. DIAGNOSTIC DIFFERENTIATION (CENTRAL VS. PERIPHERAL)



A. Cerebral Hypotonia Criteria



Abnormal Brain Functions: Presence of seizures or a disturbed conscious level.



Cognitive Deficit: Association with mental retardation.



Dysmorphic Features: Phenotypic signs of chromosomal disorders (e.g., Down syndrome).



Deep Tendon Reflexes: Characterized by hyperreflexia.



B. Lower Motor Unit Disorders Criteria



Deep Tendon Reflexes: Characterized by areflexia or hyporeflexia.



Muscle Bulk: Prominent muscle atrophy and fasciculations.



Neuropathy Pattern: Total loss of reflexes with residual movements.



Myopathy Pattern: Diminished reflexes that are directly consistent with the degree of muscle weakness.

Diagnostic Differentiation Test



These lower motor unit conditions are definitively differentiated by performing Nerve Conduction Velocity (NCV) and Electromyography (EMG).



Duchenne Muscular Dystrophy (DMD) & Muscular Dystrophies

Duchenne Muscle Dystrophy Core Facts



It is the **commonest** type of dystrophy.



Genotype: X-linked recessive (XLR) disorder.



Pathophysiology: Due to a defect in a protein called dystrophin of the skeletal muscle, cardiac muscle and the brain.

Clinical Picture



Sex: Usually male.



Age of onset: Usually 3-5 years.

Main Characteristic Weakness Features:



- Bilateral symmetrical, proximal > distal, with no sensory manifestation.



- Weakness of the shoulder girdle muscles: Unable to raise arm above the head, winging of scapula (weak serratus anterior).



- Weakness of pelvic girdle muscles: Waddling lordotic gait (weak gluteus medius and minimus), Difficulty climbing up stairs, and Positive Gower sign.



- Pseudohypertrophy of the calf muscles, Deltoid and forearm.



- Preserved muscles: Hand muscles, extraocular muscle, urethral and anal sphincter, diaphragm.

Associated Features:



- Cardiomyopathy (constant feature).



- Mental subnormality (frank MR in 25% of cases).



- Frequent respiratory infections and UTI.

Muscular Dystrophies General Definition



Primary myopathy



Progressive



Genetic basis



Characteristic muscle fiber degeneration and muscle cell death at some stage

Course & Progression



Gradually progressive; most patients are unable to walk at 12 years, and death usually occurs by the end of the second decade caused by respiratory failure or heart failure.

Diagnostic Investigations



Creatine kinase (CK) / Creatine phosphokinase (CPK): Markedly elevated.



Electromyography (EMG): Shows a myopathic pattern.



Muscle biopsy (Diagnostic): Shows muscle fiber degeneration with replacement with fat and fibrosis.



Prenatal diagnosis: Amniocentesis or chorionic villous sampling → genetic diagnosis.



Detection of female carrier: High serum creatine phosphokinase (CPK).

Treatment Lines



- No specific treatment.
- Adequate nutrition and treatment of complications.
- Treatment under trials: Gene therapy, myoblast transfer.

Becker Muscle Dystrophy Comparison Note



X-linked recessive disease of late onset during late childhood, with a slowly progressive course and a longer time of survival. It is due to dysfunctioning dystrophin or insufficient production of dystrophin.

INFECTIOUS DISEASES GUIDE: KEY FEATURES & MANAGEMENT

MEASLES (Rubeola)

Causative Organism
Measles virus



Mode of Infection
Droplet



Infectivity Period

7th day after exposure to 5th day after onset of rash

Incubation Period: 10 days



Clinical Picture

Prodromal Stage

4 days before rash

1. **Fever**
2. **Bronchitis**
3. **Conjunctivitis**
4. **Koplik spots** (Pathognomonic, buccal mucosa, opposite 2nd molars)
5. Generalized lymphoid hyperplasia



Eruptive Stage

1 days before rash

1. **Maculo-papular rash**, may be hemorrhagic (**Black measles**)
2. Starts at **hairline** & spreads downwards, to face, neck, upper limbs, then lower limbs & feet by 3rd day
3. Fades in same sequence.

Convalescence

Symptoms improve as rash disappears.



Investigations

1. Leukopenia, relative lymphocytosis
2. Rise of anti-measles Ab titer
3. **Virus isolation** from blood, urine & naso-pharyngeal secretions during infectivity



Complications

Respiratory
Pneumonia, Otitis media, Thrush, Laryngitis, etc.



Neurological
Encephalitis (post-infectious, SSPE), Multiple sclerosis, etc.



Prevention

1. **Active immunization:** living attenuated virus
2. **Passive immunization:** Immune serum globulin (within 5 days of exposure)



Treatment

1. **Isolation** till 3 days after rash
2. Bed rest, fluids, light diet
3. Antipyretics, sedatives, cough mixtures
4. **Antibiotics** if secondary infection
5. **Vitamin A** (Reduces morbidity/mortality in developing countries)



CHICKENPOX (Varicella)

Causative Organism

Varicella-zoster virus (VZV)



Definition & Characteristics

Highly contagious childhood illness. Acute infectious exanthematous disease; successive crops of **vesicles** on skin & mucous membranes; mild constitutional reaction.

Incubation Period: 2-3 weeks

Mode of Infection Droplet, or direct contact with vesicles (not crusts)

24 hours before rash, till all lesions are crusted.



Clinical Presentation

Fever & itchy **vesicular rash** in different stages ("dew drops on a rose petal"). Rash begins on **trunk** then spreads to face & limbs.



MUMPS (Epidemic Parotitis)



Causative Organism: Mumps virus (RNA paramyxovirus)

Definition & Characteristics

Acute infectious disease; **painful swelling** of salivary glands, mostly **parotids**, with constitutional symptoms. Spread by respiratory droplets or direct contact with **infected fomites**.



Clinical Picture

1. Short prodromal stage: **fever**, muscle pain, headache, malaise
2. **Parotid swelling** is bilateral (80%) or unilateral, painful, tender



Complications

Orchitis, pancreatitis, meningitis, thyroiditis, mastitis, myocarditis, arthritis, hepatitis, nephritis, unilateral deafness, ocular, Mumps Embryopathy



Treatment Protocol

Prophylactic: Active immunization (MMR vaccination)



Curative Treatment:

1. Bed rest
2. Diet: nutritious, no spices/sour fluids
3. Analgesics & antipyretics
4. **Treatment of orchitis:** bed rest, local support to testes, analgesics, systemic corticosteroids/NSAIDs for 2 weeks



MENINGITIS (Acute Bacterial / "Septic")

Definition

Acute inflammation of the **meninges** covering brain & spinal cord, with purulent exudate due to bacterial or viral infection.



Etiology by Age

- **Neonatal:** E. coli, Group-B streptococci, Listeria monocytogenes
- **Childhood:** Meningococci, Pneumococci, H. influenzae type-b (only till 5 years)



Transmission & Incubation

Droplet infection from patients/carriers.

Incubation: 3-4 days



Clinical Manifestations

Abrupt onset; high-grade **fever**, vomiting, irritability, **neck stiffness**, altered consciousness.

Infants: bulging/tense anterior fontanel, poor feeding



Features of Increased Intra-cranial Tension

- 1) Headache & Vomiting (projectile)
- 2) Bulging anterior fontanel (< 1 year)
- 4) Changes in sensorium
- 5) Focal neurological signs

Signs of Meningeal Irritation

a- **Neck Stiffness** (chin can't touch chest),

Back Stiffness

b- **Positive Kernig sign**

c- **Positive Brudzinksi's sign**

d- **Skin rash:** generalized petechial rash (meningococcal)



Diagnostic Investigations

CSF examination: Lumbar puncture MUST be done (culture, antibiotic sensitivity)

Blood culture

WBC: High, neutrophil predominance



Management Protocol

Medical emergency

IV infusion immediately after LP for antibiotics, fluids

Specific Antibiotic Treatment:

Infants < 6 months: **ampicillin + gentamicin (IV)**

Older infants & children: **IV ampicillin + IV chloramphenicol**

Supportive treatment & complications

Prevention: Chemoprophylaxis & vaccination

ENCEPHALITIS

Definition: Inflammation of brain tissue, commonly viral.



Clinical Presentation

Fever, seizures, disturbed consciousness (sensorium), confusion, focal neurological signs.



Core Syllabus Note: Can present with **Rash** and disturbed sensorium.

Etiology: HSV (Herpes Simplex Virus) is important cause in children.



Management

Requires **urgent antiviral/supportive treatment**



INFECTIOUS MONONUCLEOSIS

Causative Organism

Epstein-Barr virus (EBV)



Clinical Presentation

Fever, sore throat, cervical lymphadenopathy, and fatigue



Hepatosplenomegaly may occur

Peripheral Blood Findings: Shows atypical lymphocytes



Core Syllabus Note: Caused by Epstein-Barr virus with atypical lymphocytes

Diagnosis: Confirmed by **Monospot test** or EBV serology



ROSEOLA INFANTUM



Causative Organism: Human herpes virus-6 (HHV-6) mainly

Epidemiology: Common in infants aged **6 months-2 years**

Clinical Presentation: Characterized by **high fever** for

3-5 days, followed by **sudden fall of temperature** and appearance of **maculopapular rash**



Core Syllabus Note: Maculopapular rashes appeared with onset of **drop of fever**

Rash Distribution: Rash usually starts on the **trunk** then spreads to the neck and limbs

Complications: **Febrile convulsions** may occur

